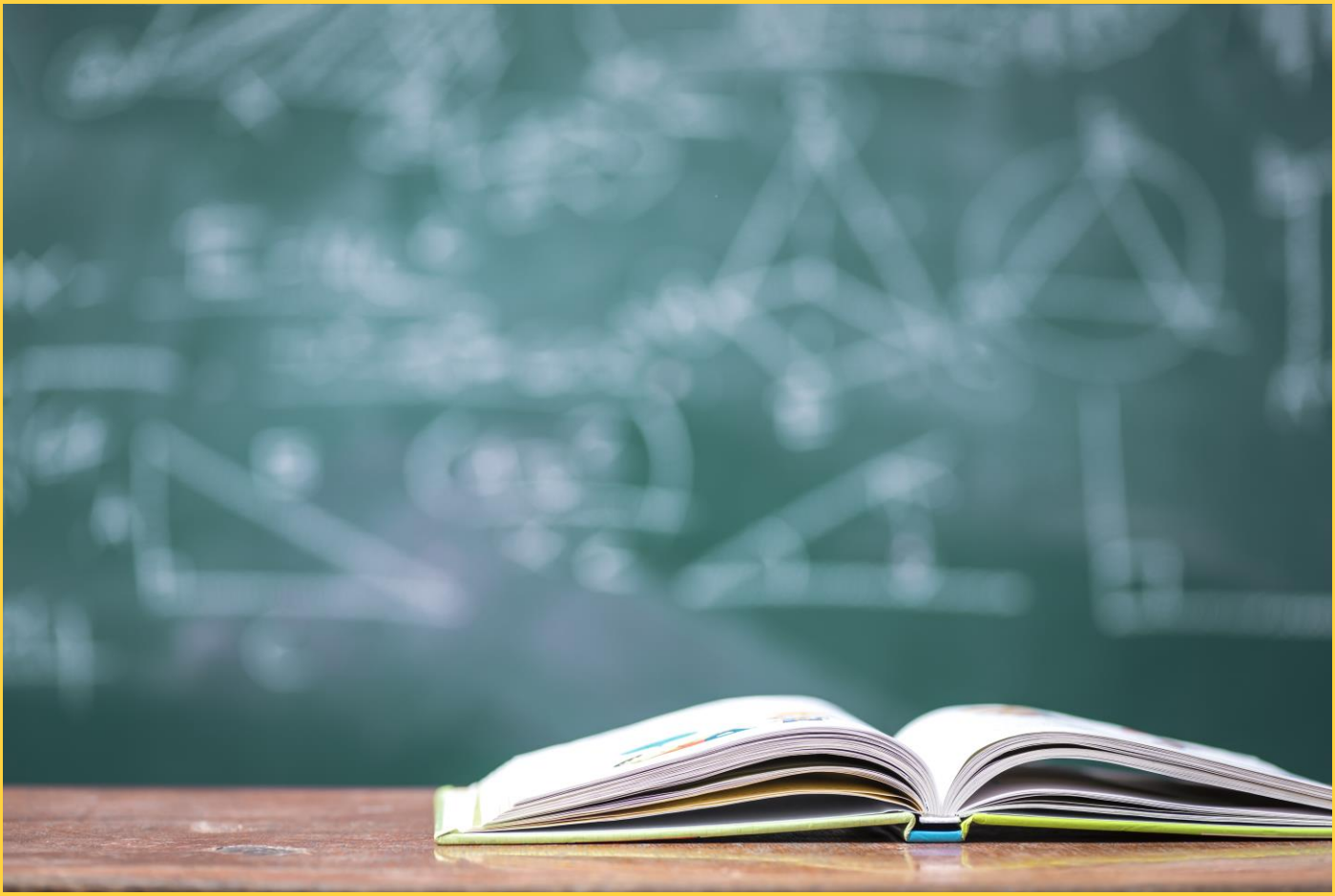

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How Education Drives Economic Growth

Prepared for:

Semester Project

Systematic Methods

INFX 502

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DATA SET

Description of the dataset

I found a dataset by Omar Mohamed that looks at how education levels connect to economic performance in different countries and continents. It brings together numbers like literacy rates, how many doctors there are per person, GDP per person, and unemployment rates. The goal is to see if there are any patterns and to figure out how investing in people's education might affect a country's economy. This dataset was uploaded in August 2025, so it is pretty recent. You can find it at https://www.kaggle.com/omarmohammed.

This dataset stood out to me when I was looking through Kaggle. I have always wondered if education really makes a difference in how well a country does. When I was younger, I wanted to get as much education as I could so I could support myself. Lately, though, I have talked to some younger family members who think college is not worth it. Even though this dataset looks at a much bigger picture than just my own experience, I am interested to dig into the numbers and see what I can learn.

Variables names and their descriptions

This data set has information on 191 countries and covers a mix of socioeconomic and health factors. Some of the main numbers in the data are literacy rate, physician density, total and per capita GDP, GDP growth rate, and unemployment rate. These numbers vary a lot from country to country, which you can see in the table below. There are three main categories in the data: Country, GDP per Capita Category, and Continent. I labeled everything in the table and added a short description for each. I also changed the name of the data set from education_economy_data to edu_money to make my code easier to write. I think the original name was a better fit and less biased, but I went with the shorter name to keep things simple.

Variable name	Class	Range	Disruption
Country	character	191 unique entries	Country names
Literacy Rate	numeric	26.8 to 100.0	Indicates the percentage of the population who can read and write. A key indicator of educational development and human capital.
Physician Density	numeric	0.01 to 8.42	Number of physicians per 1,000 people. Reflects access to healthcare services and medical infrastructure.
GDP Current USD	numeric	6.228031e+07 to 2.772070e+13	Total gross domestic product in current U.S. dollars. Measures the overall economic output of a country.

GDP Growth (% Annual)	numeric	-20.11158 to 33.80262	Annual percentage change in GDP.
GDP per Capita Category	numeric	193.0071 to 256580.5151	GDP is divided by population. Reflects average economic productivity and standard of living per person
GDP per Capita Category	character	4 Categories Very high to very low	Categorizes countries based on GDP per capita
Unemployment rate	numeric	0.13 to 50.00	Percentage of the labor force that is unemployed
continent	character	6 categories "Africa, Asia, Erupe, north America, Oceania, and South America"	Geographic regions of each country.

Head of the dataset

The following table is what is printed when the code >head(education_economy_data) is executed. (Note: It is reformatted in a table to be easy to read.)

	Country	Literacy Rate	Physician Density	GDP Current USD	GDP Growth (% Annual)	GDP per Capita Category	GDP per Capita Category	Unemployment rate	continent
1	Afghanistan	43.0	0.28	1.723305e+10	2.71088679	415.7074	Very Low	13.991	Asia
2	Albania	98.1	1.20	2.354718e+10	3.93662526	8575.1711	Mid	10.108	Europe
3	Algeria	81.4	1.72	2.476260e+11	4.10000000	5364.0279	Low	11.701	Africa
4	Andorra	100.0	3.33	3.785067e+09	2.58355481	46812.4485	High	3.700	Europe
5	Angola	71.1	0.21	8.482465e+10	1.00128936	2308.1598	Low	14.537	Africa
6	Antigua and Barbuda	99.0	2.76	2.033085e+09	3.86201219	21787.1017	High	7.700	North America

Expectations during analysis

Before I started looking at the data, I expected that countries with higher literacy rates would have loosely stronger economies. I thought this would show up as higher GDP, faster GDP growth, and more GDP per person. So, all these factors would be strongly linked. I also figured that countries with more doctors, which usually means better access to healthcare, would do better in these areas too. My thinking was that when a country invests in its people through education and healthcare, it helps the whole country grow. On the other hand, if people value education, they tend to have higher incomes, which boosts GDP growth and increases GDP per capita. Education gives people skills that make them more productive, lowers unemployment, and helps the economy get stronger.

In the end, both ways of looking at it, whether you focus on investing in people or on how much people value education, probably lead to the same result. Both would show how important education is for a country's economic success.

- **Hypothesis A:** Higher literacy rates are associated with higher GDP per capita.
- **Hypothesis B:** GDP, GDP growth, and GDP per person are all correlated.
- **Hypothesis C:** Countries with higher physician density have lower unemployment rates.
- **Hypothesis D:** Countries in the “Very High” GDP per capita category have significantly lower unemployment rates than those in the “Very Low” category.

ANALYSIS

CORRELATION

I decided to start by looking at a correlation chart to see which groups in the data set are most closely related. My main goal here was to figure out where I should focus first. At first, I thought that GDP, GDP growth, and GDP per capita would be the most strongly correlated. But I was surprised to see that these variables only had weak or even weak negative correlations.

Looking at the table, I noticed that variables like literacy rate, physician density, and GDP per capita are all positively correlated. Physician density has a strong correlation with GDP per capita (0.54), which could mean that countries with more wealth tend to have better access to healthcare. Literacy rate also shows a moderate correlation with GDP per capita (0.37) and a strong one with physician density (0.61). On the other hand, GDP growth doesn't seem to be strongly connected to most of the other indicators. The unemployment rate is negatively correlated with all the other variables, especially GDP per capita (−0.20) and physician density (−0.17). This makes me think that better economic and health conditions might be linked to a lower unemployment rate.

```
> edu_money.sub<- data.frame(edu_money$`Literacy Rate`, edu_money$GDP, +
        edu_money$`Physician Density`, edu_money$`GDP Growth`+
edu_money$`GDP per Capita`, edu_money$`Unemployment Rate`)
> cor(edu_money.sub)
```

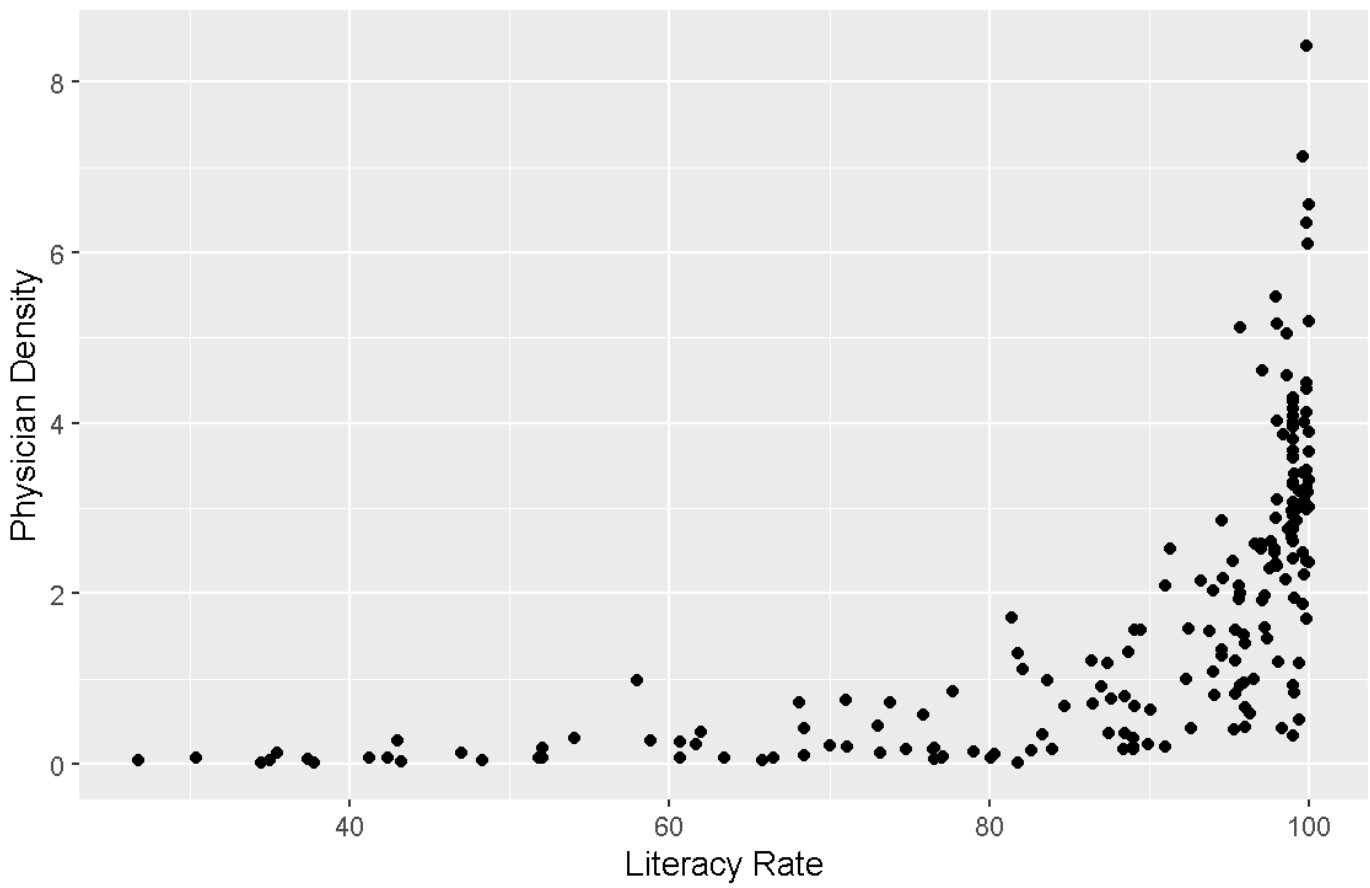
	Literacy Rate	Physician Density	GDP	GDP Growth	GDP per capita	Unemployment Rate
Literacy Rate	1.000000000	0.61015708	0.1186403	-0.02234683	0.36640190	-0.005537313
Physician Density	0.61015708	1.00000000	0.09514566	-0.12672799	0.54078147	-0.17269354
GDP	0.118640376	0.09514566	1.00000000	0.00538374	0.17623273	-0.090067179
GPD growth	-0.022346829	-0.12672799	0.00538374	1.00000000	-0.09611304	-0.089854880
GPD per capita	0.366401895	0.54078147	0.17623273	-0.09611304	1.00000000	-0.203203162
Unemployment Rate	-0.005537313	-0.17269354	-0.09006718	-0.08985488	-0.20320316	1.000000000

SCATTER PLOTS

Literacy Rate x Physician Density

The graph shows that as literacy rates go up, physician density usually goes up too. When literacy rates are between 0 and 60, physician density stays about the same. From 60 to 90, it starts to rise, and then it goes up quickly between 90 and 100. Most of the data points are grouped around literacy rates of 98 to 100, where physician density is between 2.5 and 4.5. There are a few outliers with physician density above 6, which will show up in the box plots.

```
>                                                    ggplot(edu_money)+
+  geom_point(aes(x = `Literacy Rate`,
+  y = `Physician Density`))
```

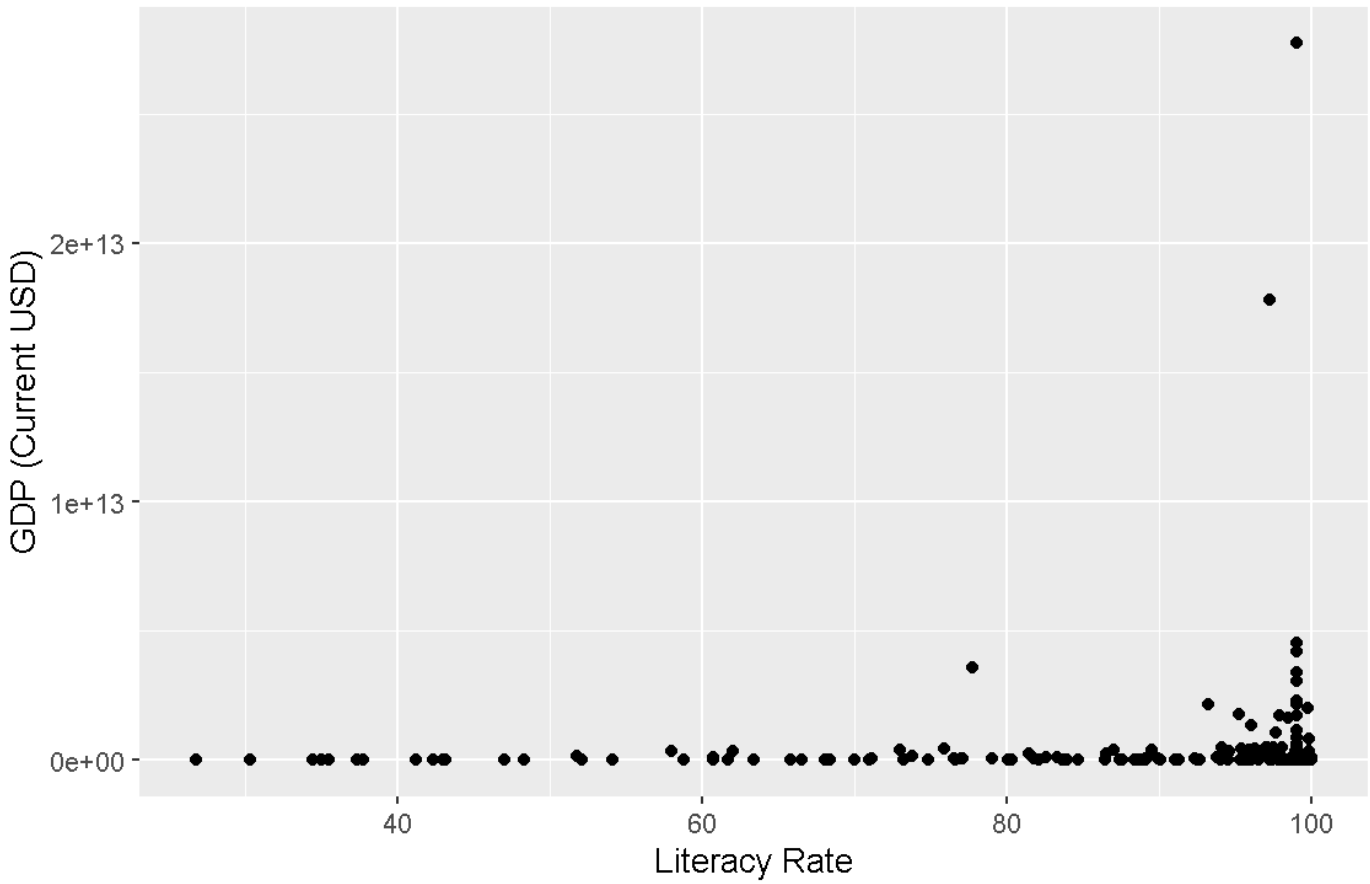


SCATTER PLOTS

Literacy Rate x GDP (Current USD)

This graph shows that GDP does not change until literacy rates are clustered between 90 and 100 percent. In this high literacy range, many countries are represented, but their GDP values vary widely. Most countries in this group have lower GDP, while a few are extreme outliers with much higher GDP, indicating no clear direct relationship between literacy rate and GDP in this range.

```
> ggplot(edu_money)+
+  geom_point(aes(x = `Literacy Rate`, y = `GDP (Current USD)`))
```

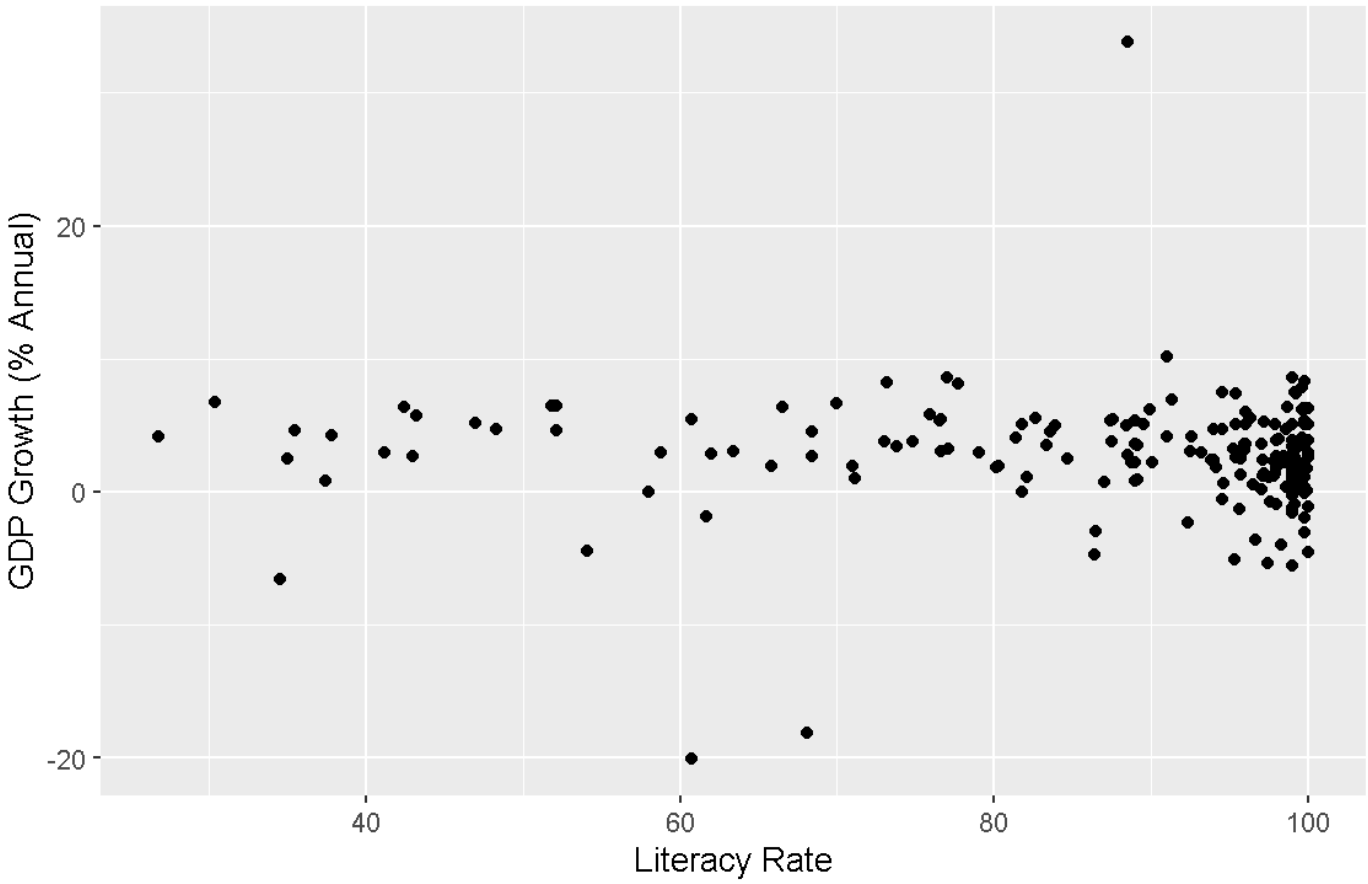


SCATTER PLOTS

Literacy Rate x GDP Growth (% Annual)

There is no visible correlation between literacy rate and GDP growth. Even countries with nearly 100 percent literacy have both positive and negative growth rates. Notable outliers include one country with approximately 95 percent literacy and GDP growth exceeding 25 percent, alongside a few others experiencing GDP growth of around -20 percent. It's noted that more countries experience growth than losses, and there is a cluster around a 100 percent literacy rate.

```
> ggplot(edu_money)+
+  geom_point(aes(x = `Literacy Rate`, y = `GDP Growth (% Annual)`))
```

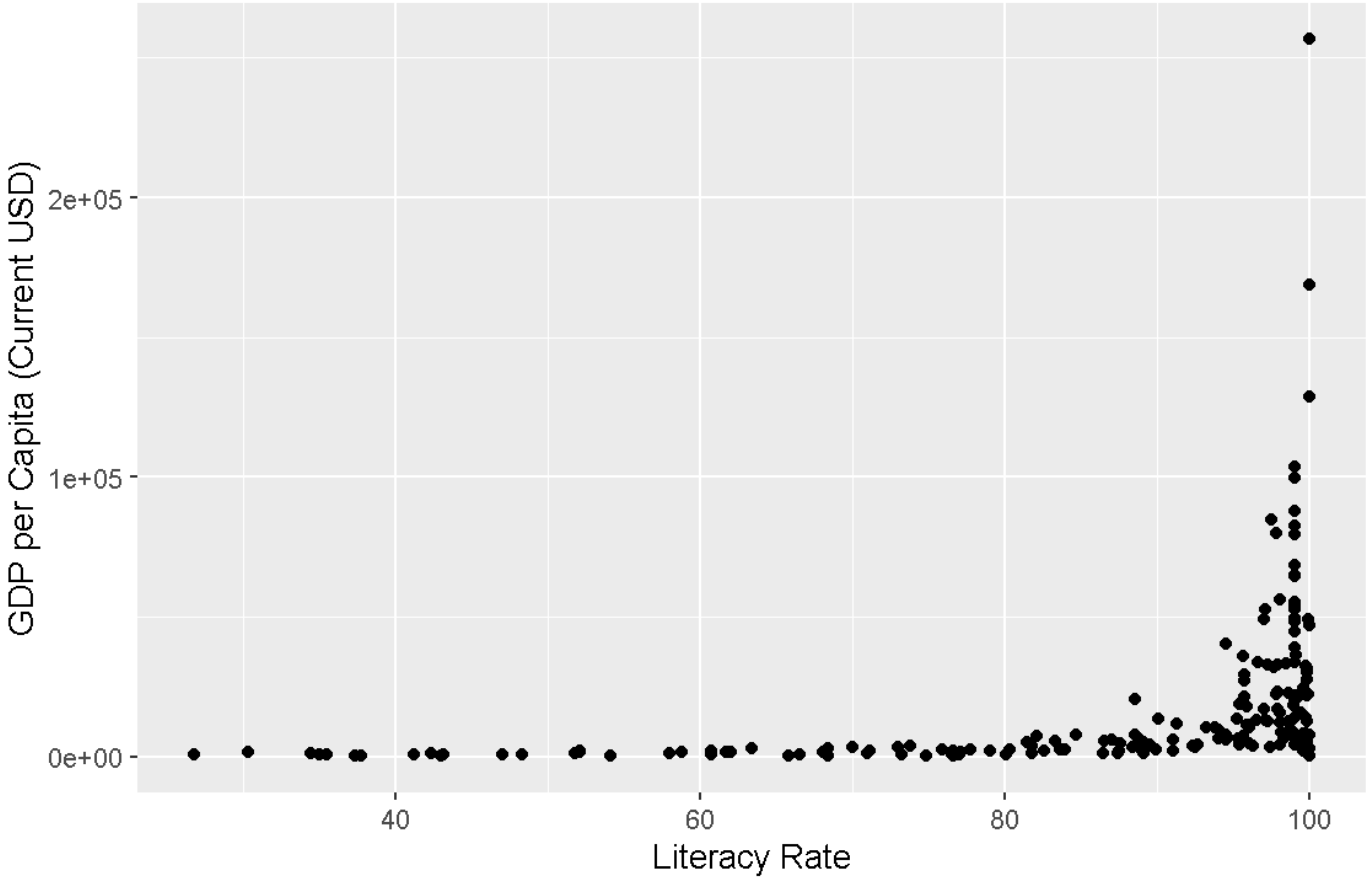


SCATTER PLOTS

Literacy Rate x GDP per Capita (Current USD)

There is not a strong linear correlation across the entire range, but a noticeable trend is present. Countries with literacy rates under 80 percent tend to have low GDP per capita. With a noticeable change in GDP per capita at an 80 percent literacy rate. Higher GDP per capita are almost exclusively associated with literacy rates from 95 to 100 percent, with some outliers that have extremely high GDP per capita and 100 percent literacy rates.

```
> ggplot(edu_money)+
+  geom_point(aes(x = `Literacy Rate`, y = `GDP per Capita (Current USD)`))
```

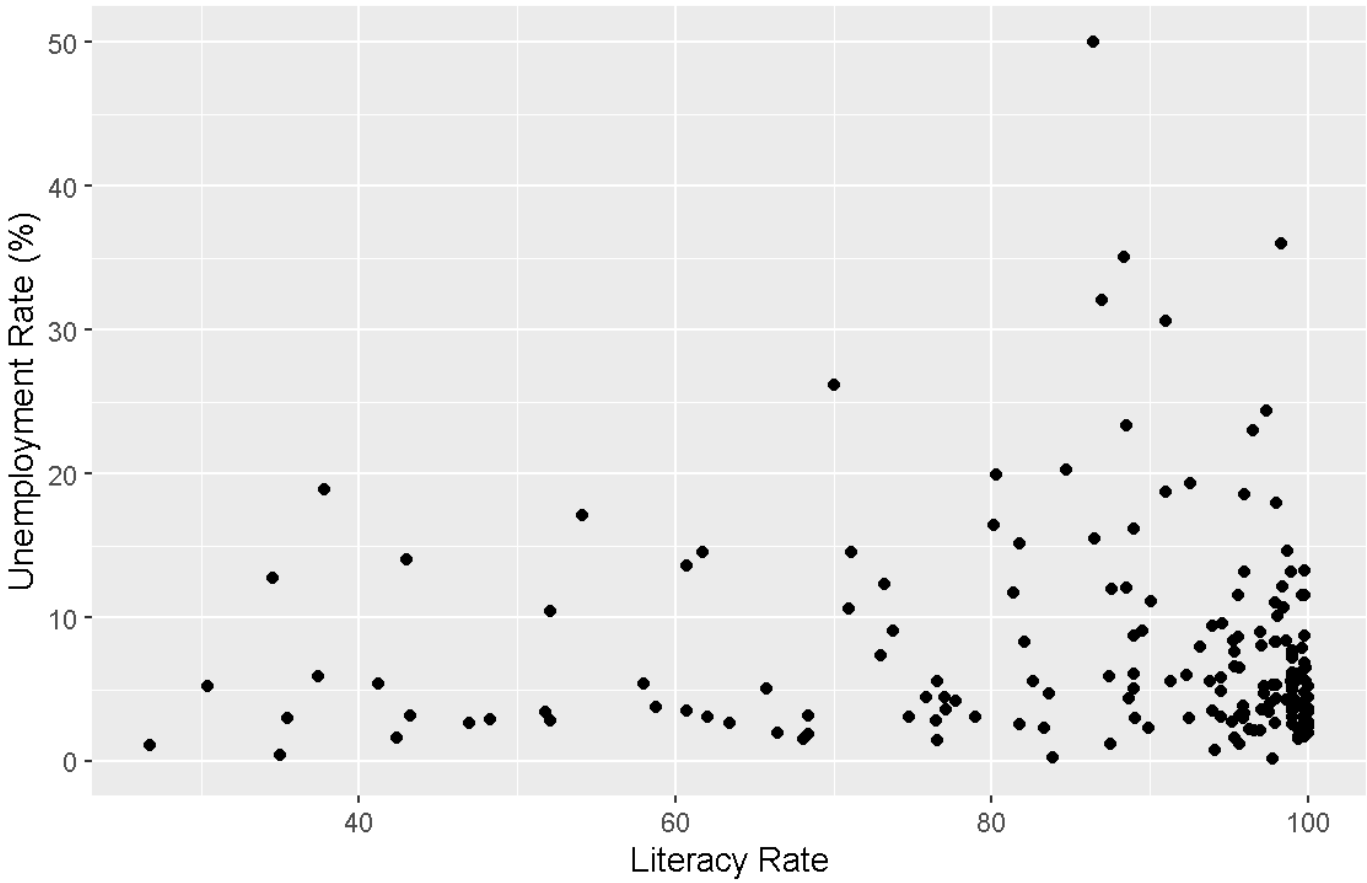


SCATTER PLOTS

Literacy Rate x Unemployment Rate

There doesn't appear to be a strong negative correlation, meaning that higher literacy rates are associated with lower unemployment rates. Instead, the data just looks scattered. Especially at high literacy rates, unemployment varies significantly from near 0 percent up to 50 percent. On the other hand, even at low literacy rates under 60 percent, unemployment seems mostly low, often under 20 percent. Additionally, a dense cluster can be seen where literacy rates are around 90 to 100 percent and unemployment is between 0 and 15 percent.

```
> ggplot(edu_money)+
+  geom_point(aes(x = `Literacy Rate`, y = `Unemployment Rate (%)`))
```

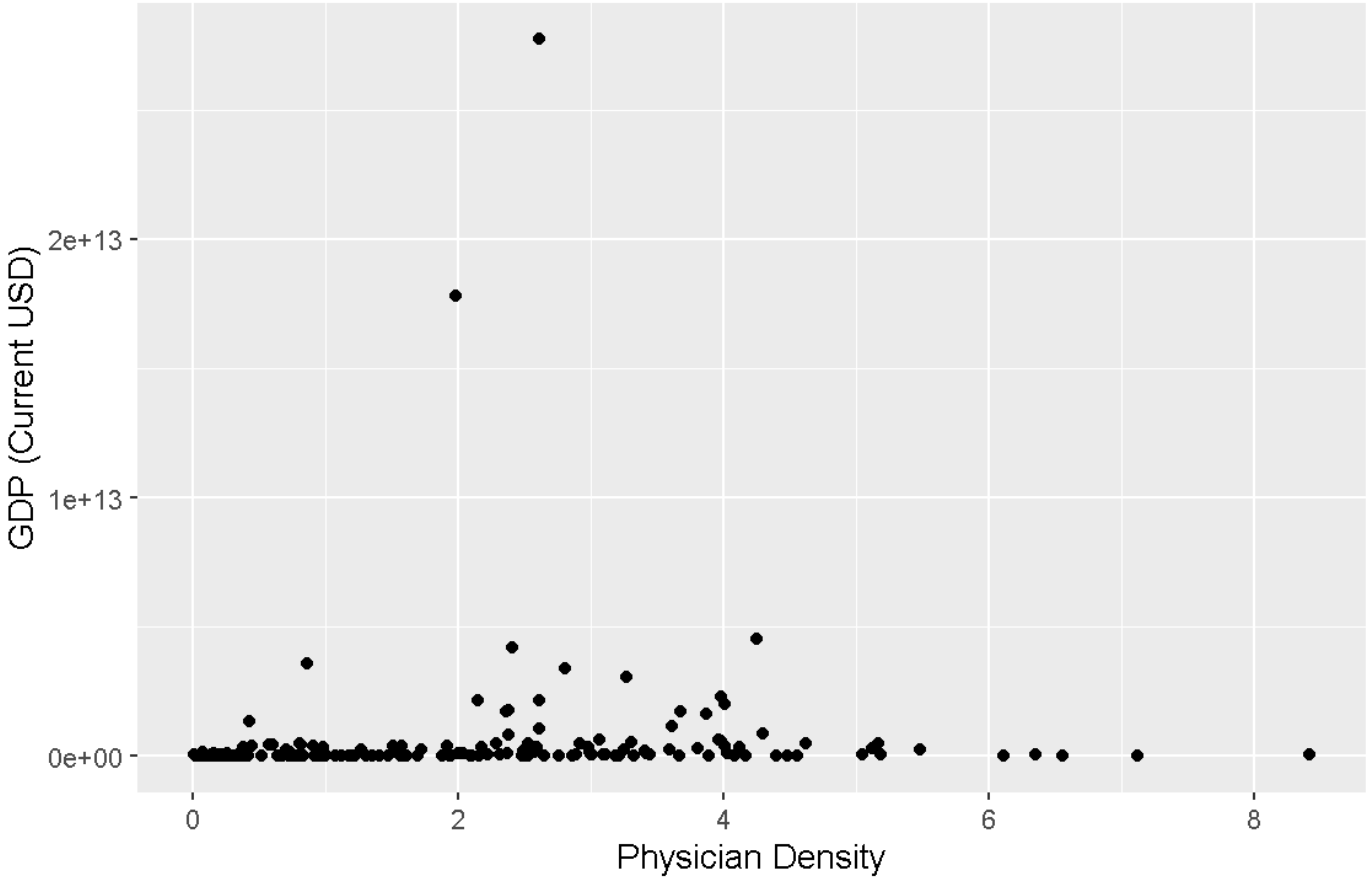


SCATTER PLOTS

Physician Density x GDP (Current USD)

There doesn't appear to be a strong linear correlation between physician density and GDP. Countries with high GDP don't necessarily have the highest physician density. There are a few extreme GDP values with physician densities between 2 and 3. All in all, it appears that GDP may not be a significant driver of physician density.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`Physician Density`, `GDP (Current USD)`))
```

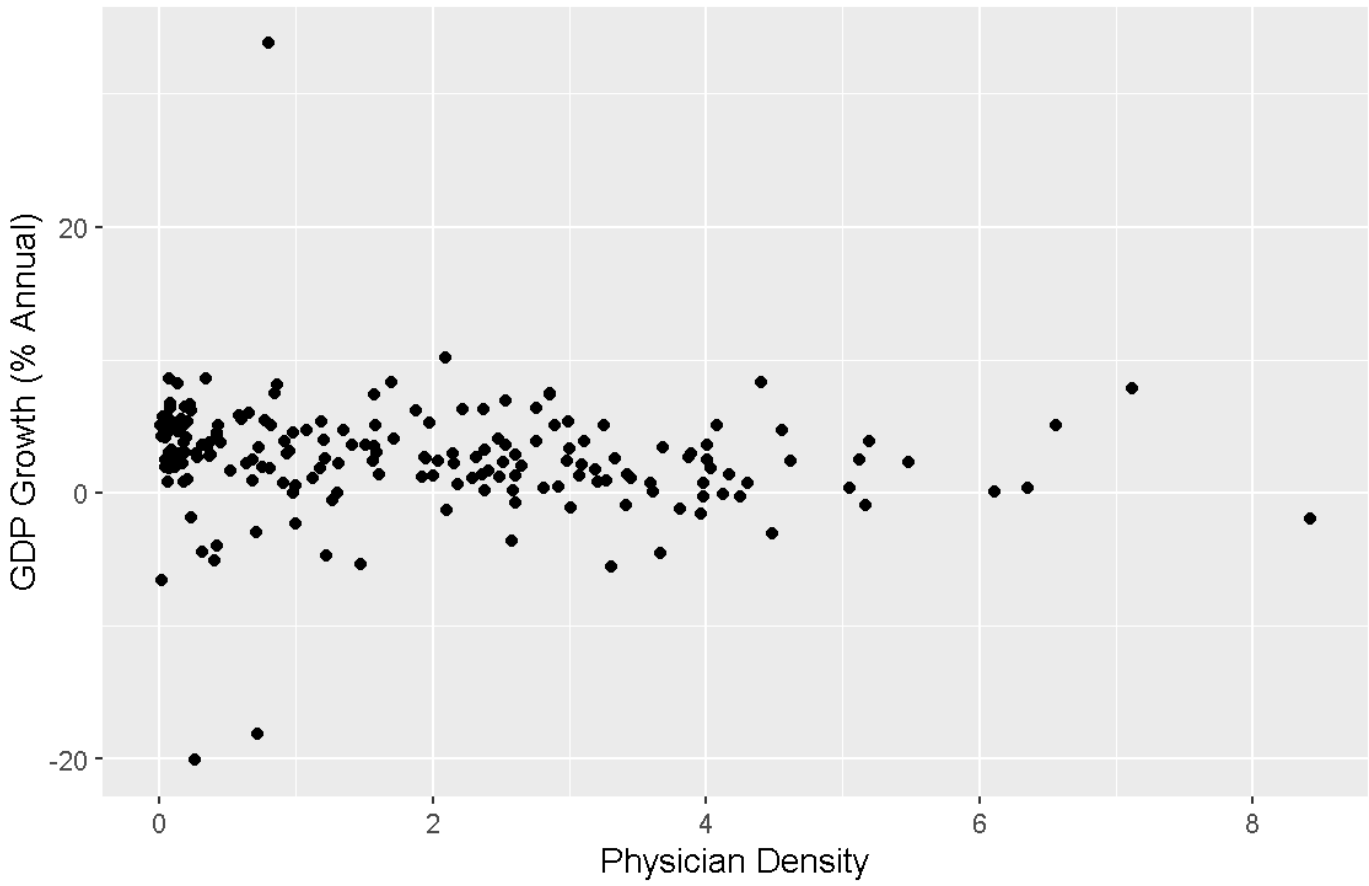


SCATTER PLOTS

Physician Density x GDP Growth (% Annual)

There does not appear to be a strong correlation between physician density and GDP growth. The points are scattered horizontally, suggesting that increasing GDP growth does not consistently lead to higher physician density. Many points fall between 0 and 4 for physician density and between -5 percent and 10 percent for GDP growth. There are a few outliers, with one point that has a GDP growth above 30 percent and a low physician density. Two other outliers show GDP growth around -20 percent, each with physician density between 0 and 1.

```
ggplot(edu_money)+
+  geom_point(aes(x =`Physician Density`, y= `GDP Growth (% Annual)`))
```

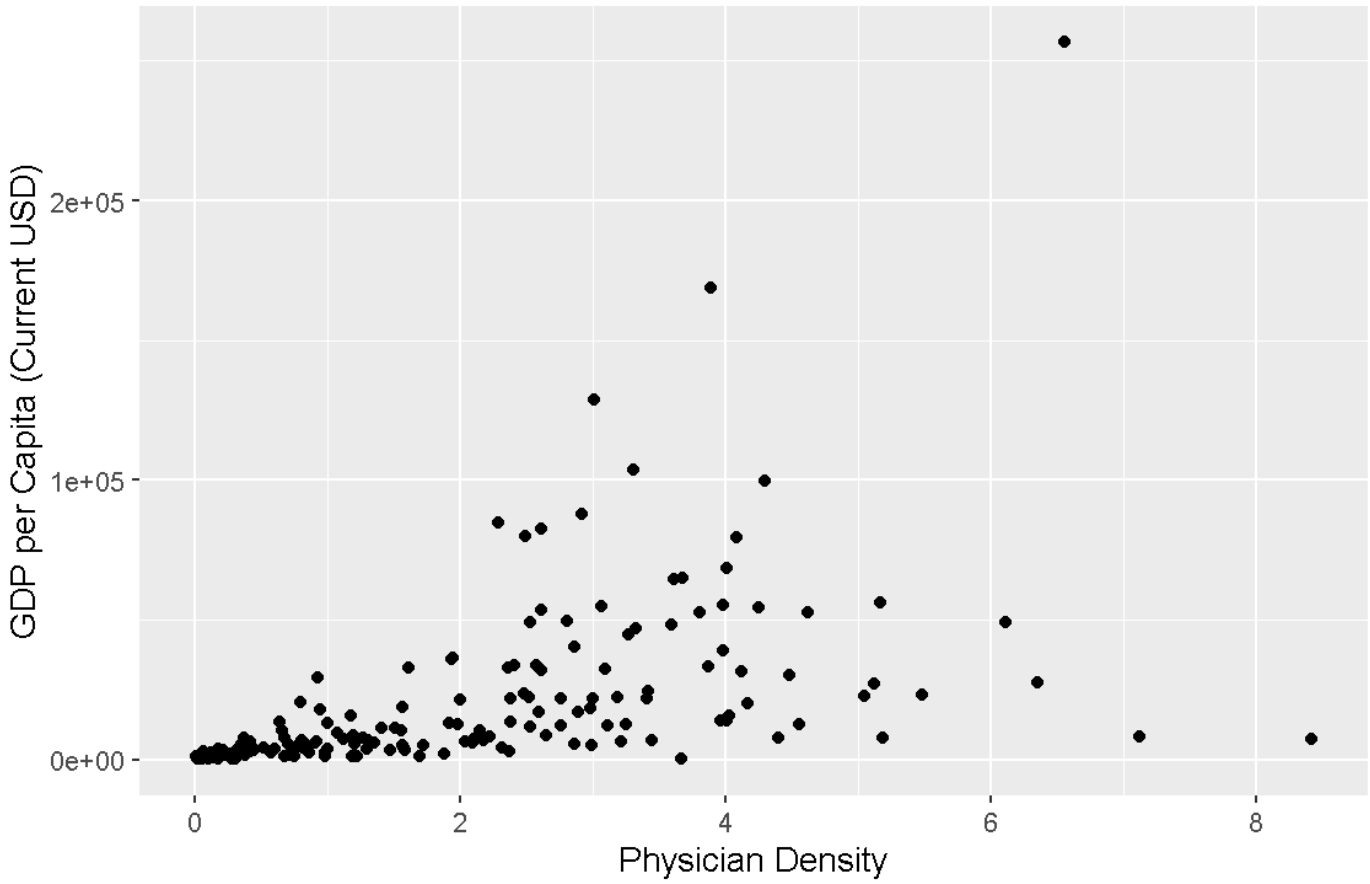


SCATTER PLOTS

Physician Density x GDP per Capita (Current USD)

Countries with higher GDP per capita generally also have higher physician density, but the correlation is not perfect. Most countries have physician densities between 0 and 4 and GDP per capita below 100,000 USD. There is one clear outlier with a physician density of 6.5 and a GDP per capita of 220,000 USD. A few others stand out with GDP above 100,000 USD, but physician density is around 3 to 4.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`Physician Density`, y= `GDP per Capita (Current USD)`))
```

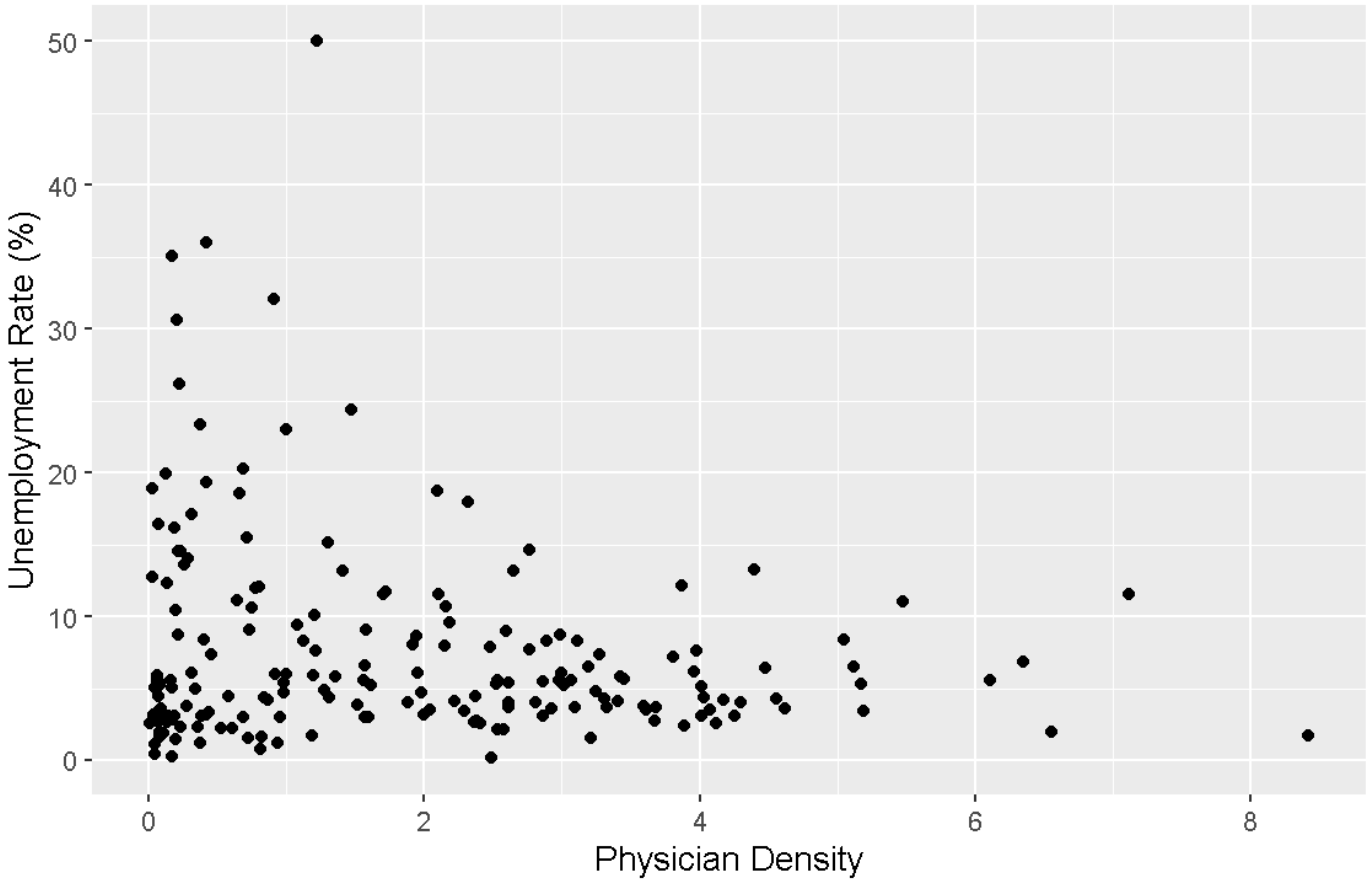


SCATTER PLOTS

Physician Density x GDP per Unemployment Rate (%)

There appears to be a negative correlation between physician density and unemployment rate. In other words, as unemployment rates decrease, physician density increases. Most points on the plot are concentrated at low physician density of 0 to 5 and unemployment rates between 0 and 20 percent. However, there are a few outliers, with one having a very high unemployment rate of about 50 percent paired with a lower physician density. Others show higher physician density above 6, but their unemployment rates are generally under 10 percent.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`Physician Density`, y= `Unemployment Rate (%)`))
```

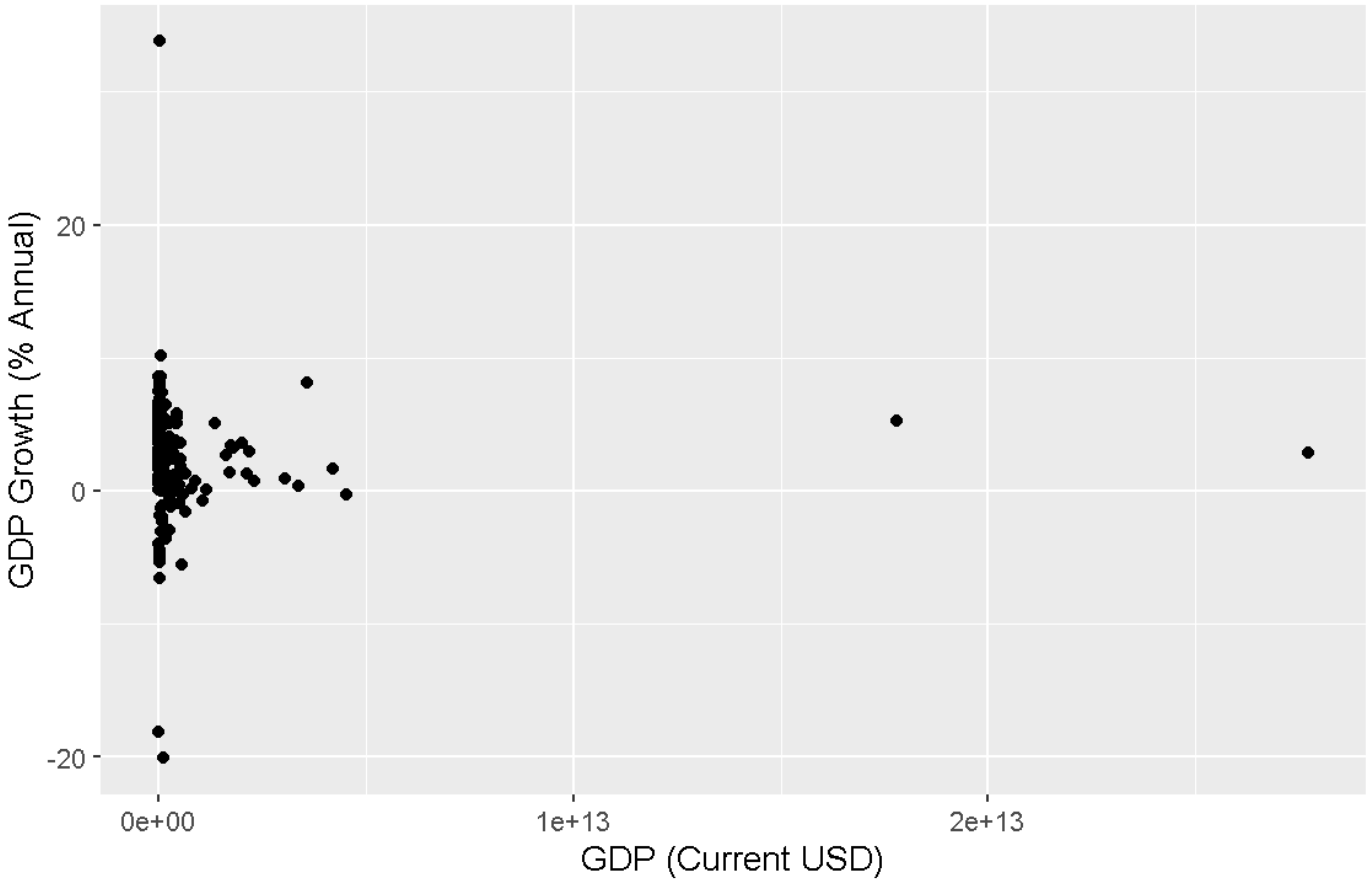


SCATTER PLOTS

GDP (Current USD) x GDP Growth (% Annual)

Most data points are near the left side of the graph, which shows a low GDP, indicating that many countries have relatively small economies compared to the largest one. Outliers are easy to find, there are two points far to the right with very high GDP, and their growth rates are around 2 to 5 percent. Additionally, countries with low GDP show a large spread in growth rates, ranging from -20 percent to 30 percent, suggesting that smaller economies experience more changes. Finally, some countries have negative GDP growth and are below the zero line.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`GDP (Current USD)`, y = `GDP Growth (% Annual)`))
```

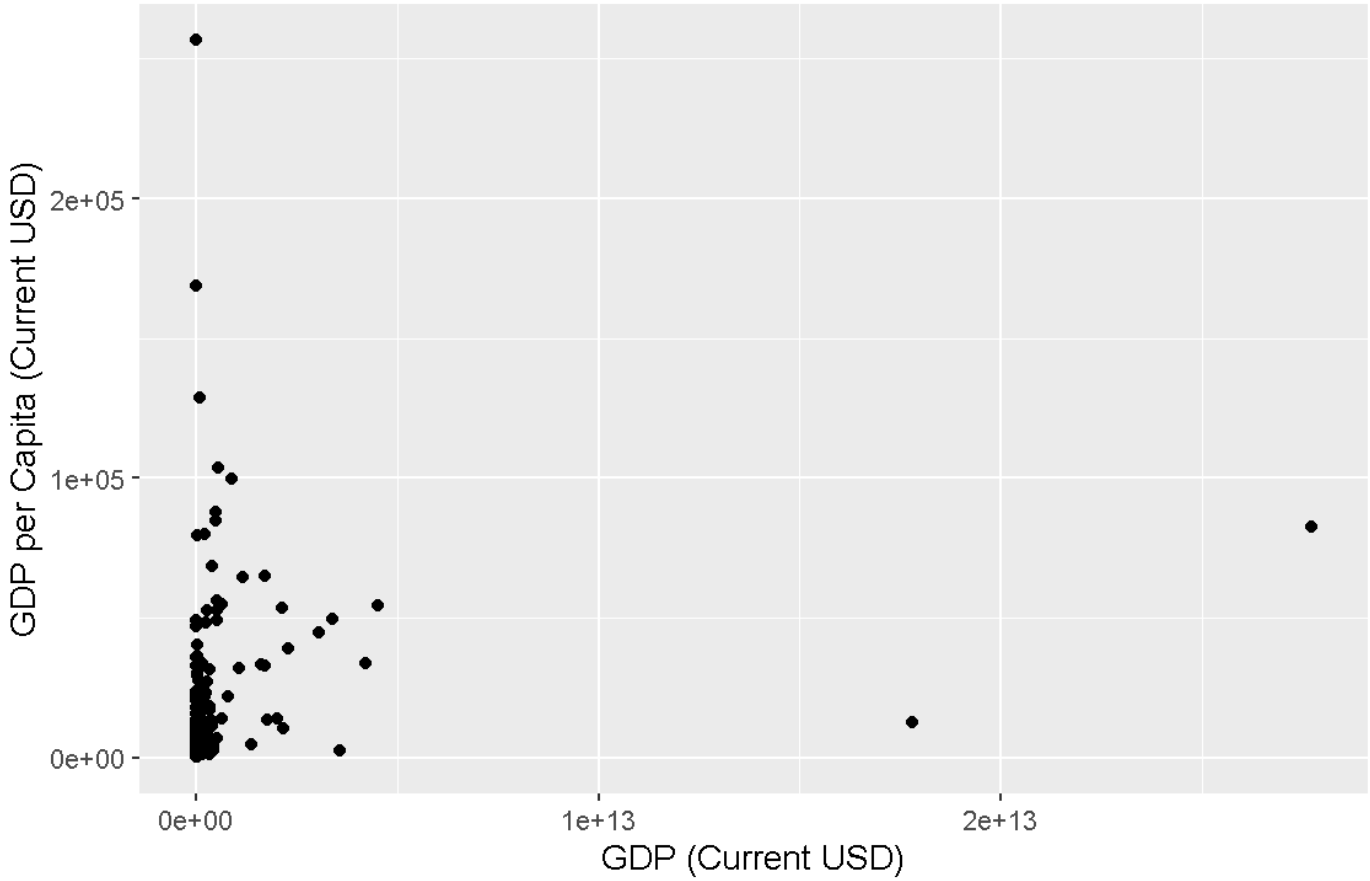


SCATTER PLOTS

GDP (Current USD) x GDP per Capita (Current USD)

There is no clear linear relationship. The plot suggests that having a large GDP does not guarantee a high GDP per capita. There are a few possible outlier points far to the right, with high GDP but not super high GDP per capita. Some of the outlier points are very high GDP per capita, but low GDPs.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`GDP (Current USD)`, y = `GDP per Capita (Current USD)`))
```

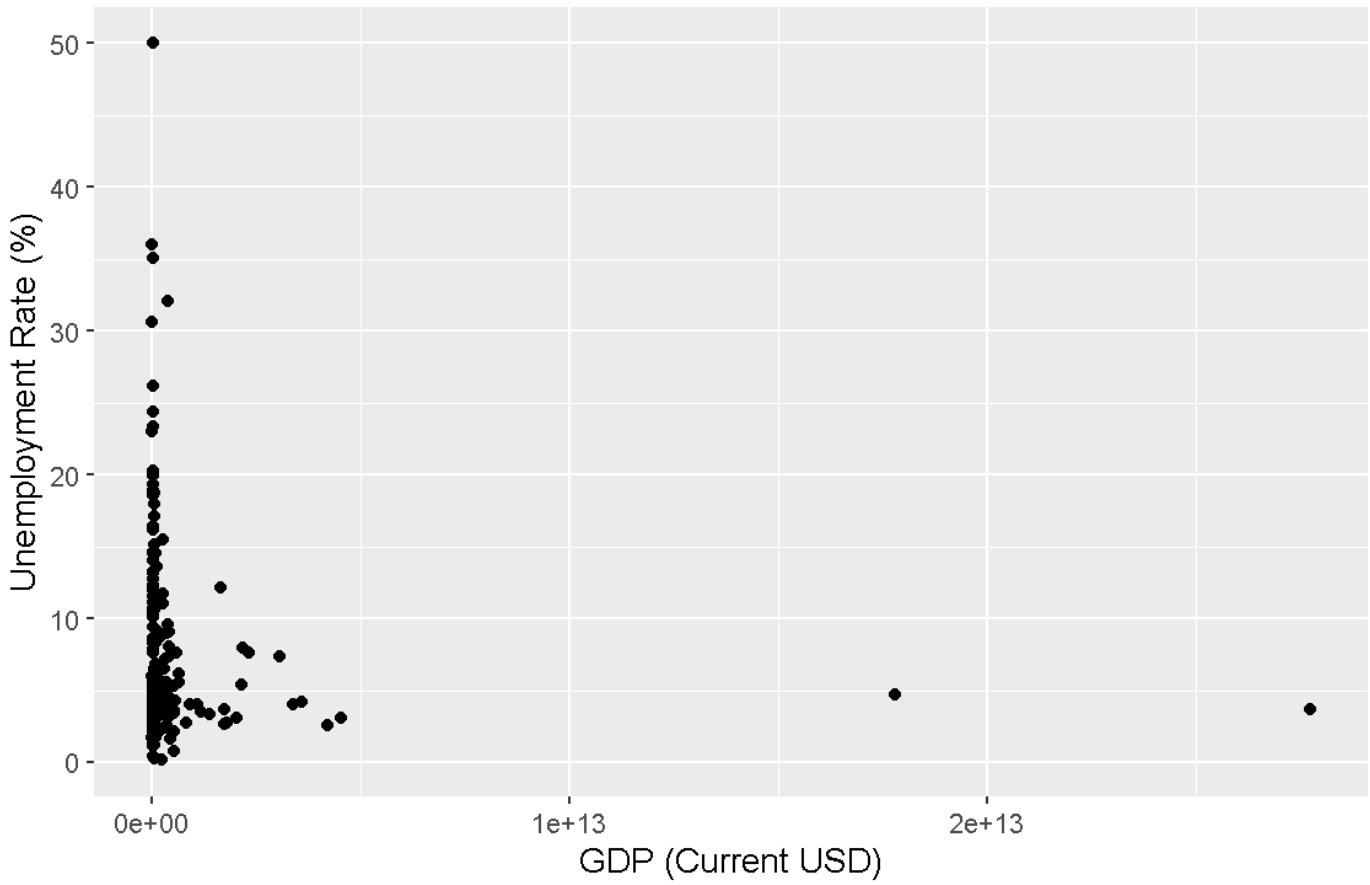


SCATTER PLOTS

GDP (Current USD) x Unemployment Rate (%)

Most of the data points are clustered very close to the y-axis because of low GDP values and varying unemployment rates. A few points are far to the right, representing countries with a very high GDP, and they all have lower unemployment rates. There is no clear linear correlation, but it does hint that countries with higher GDP tend to have lower unemployment rates. On the other hand, Countries with low GDP show a wide spread of unemployment rates.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`GDP (Current USD)`, y = `Unemployment Rate (%)`))
```

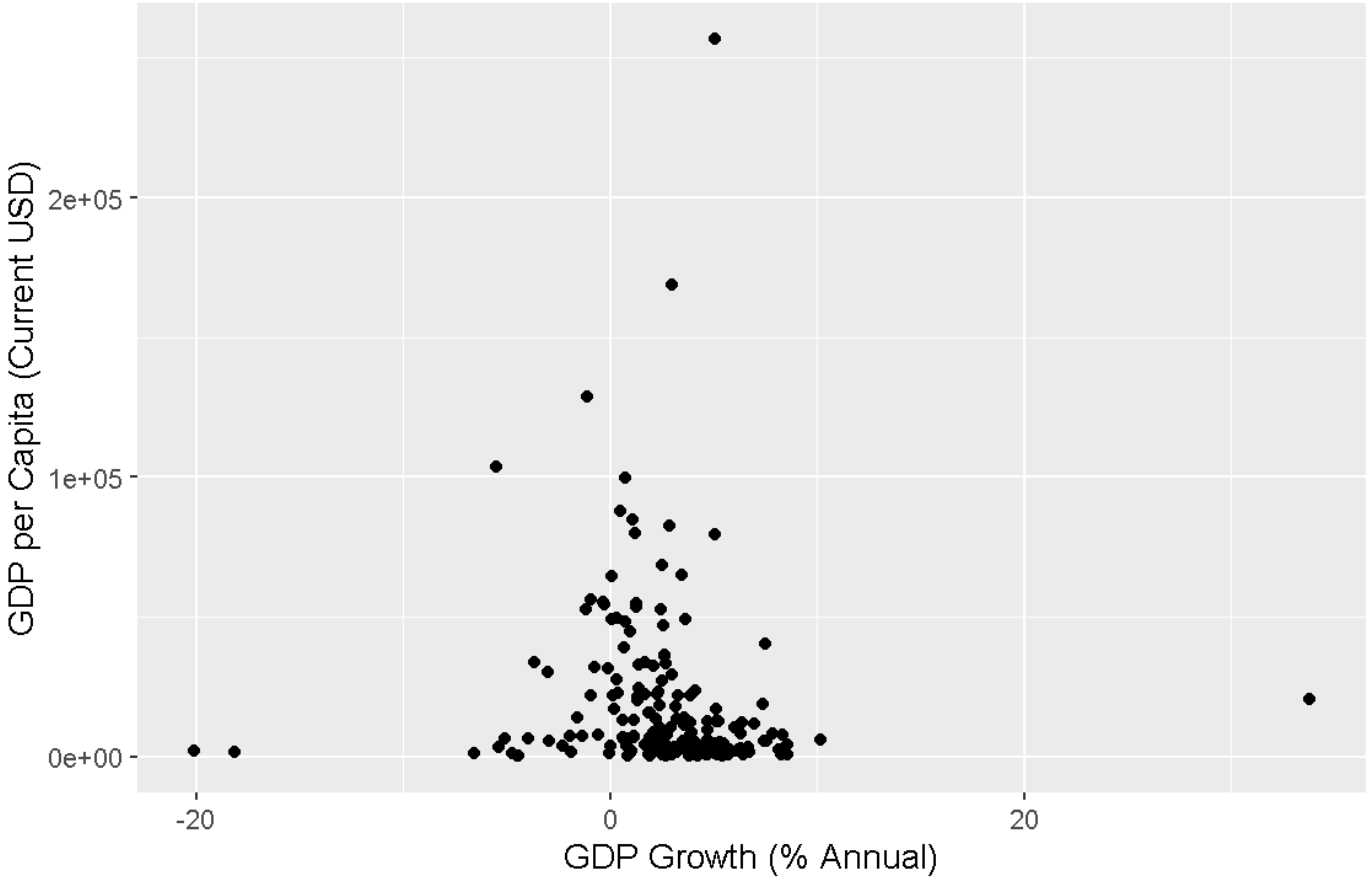


SCATTER PLOTS

GDP Growth (% Annual) x GDP per Capita (Current USD)

The data does not show a strong relationship between GDP growth and GDP per capita. High-income countries do not necessarily have high growth rates. While GDP growth rates cluster near zero, GDP per capita varies from 0 to over 100,000. This suggests that countries with similar growth rates can have different income levels. There are a few countries with GDP per capita greater than 100,000, and these countries experience little to no change in their GDP growth rates.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`GDP Growth (% Annual)`, y = `GDP per Capita (Current USD)`))
```

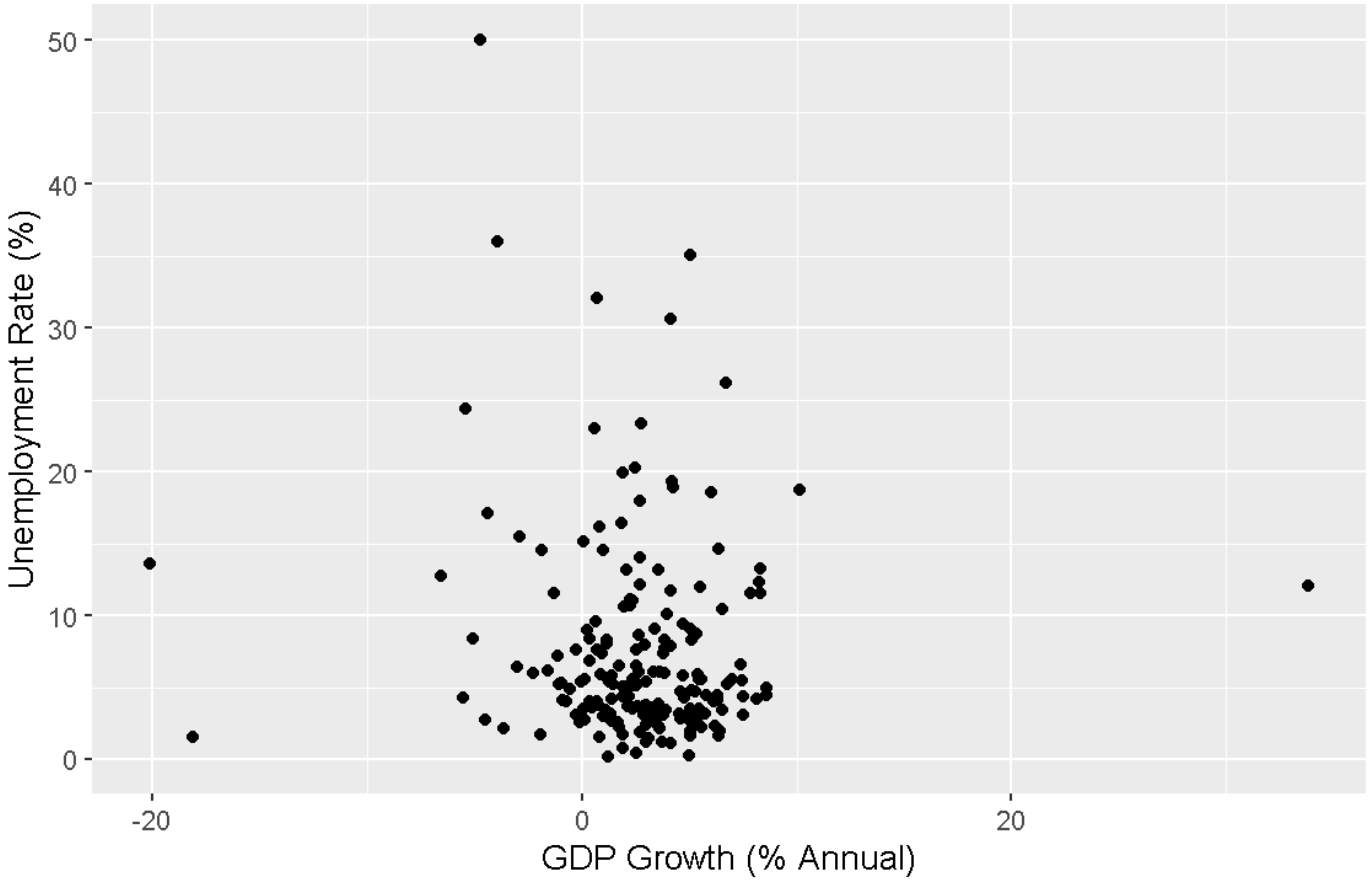


SCATTER PLOTS

GDP Growth (% Annual) x Unemployment Rate (%)

There doesn't appear to be a strong linear relationship between GDP growth and unemployment. Most points cluster near 0 to 10 percent GDP growth and 0 to 15 percent unemployment. The cluster suggests that even with positive GDP growth, unemployment can vary widely. Outliers indicate unusual economic conditions. Extreme outliers to note includes one point with GDP growth near -20 percent and unemployment around 15 percent, another with unemployment above 50 percent and GDP growth near 0 percent, and a point with GDP growth above 20 percent and unemployment around 12 percent.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`GDP Growth (% Annual)`, y = `Unemployment Rate (%)`))
```

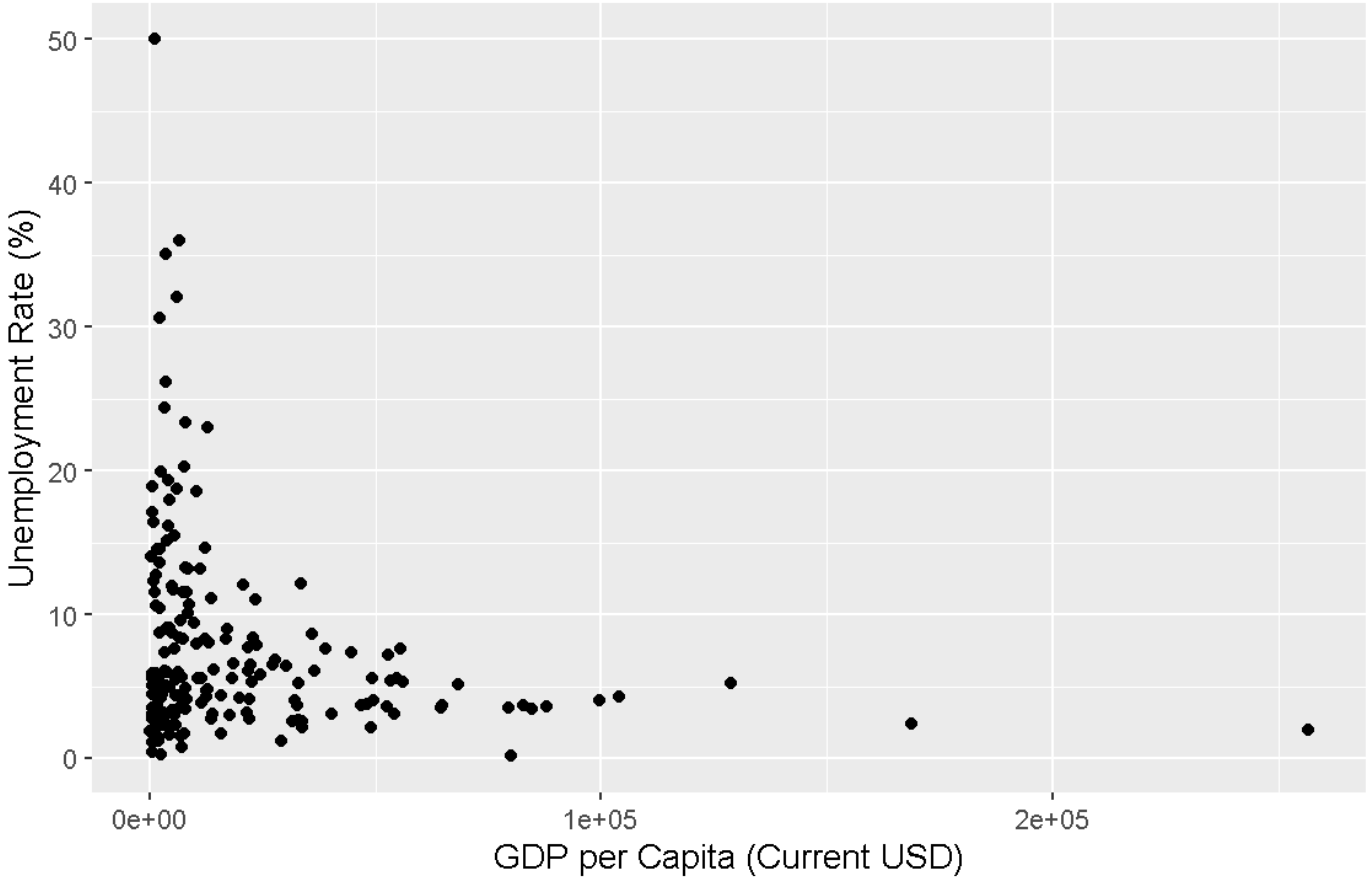


SCATTER PLOTS

GDP per Capita (Current USD) x Unemployment Rate (%)

Most points cluster near the lower end of GDP per capita (close to zero) and have varying unemployment rates. Notably, there is one extreme outlier with nearly 50 percent unemployment and a very low GDP per capita. As GDP per capita increases, however, unemployment rates tend to be mostly under 10 percent.

```
> ggplot(edu_money)+
+  geom_point(aes(x =`GDP per Capita (Current USD)`, y = `Unemployment Rate (%)`))
```



Contingency tables

- Africa has the highest number of countries in the Very Low and Low GDP categories, with no countries in the High or Very High categories.
- Europe is the opposite, it has no countries in the Very Low category and a strong presence in the High and Very High categories.
- Asia shows a balanced distribution, with countries spread across all categories.
- North America and Oceania have relatively few countries, but a mix across categories.
- South America is mostly in the Low to Mid-range, with no countries in the Very High categories.

```
> cont.table<- table(edu_money$Continent,edu_money$`GDP per Capita Category`)
> cont.table
```

	Very low	Low	Mid	High	Very high
Africa	30	18	4	0	0
Asia	10	15	10	9	3
Europe	0	2	9	18	14
North America	1	5	10	5	2
Oceania	0	10	2	1	1
South America	0	5	5	2	0

- Africa has the most amount with 52 countries.
- Oceania and South America have fewer countries, which may limit their representation across GDP categories.

```
>table(edu_money$Continent)
```

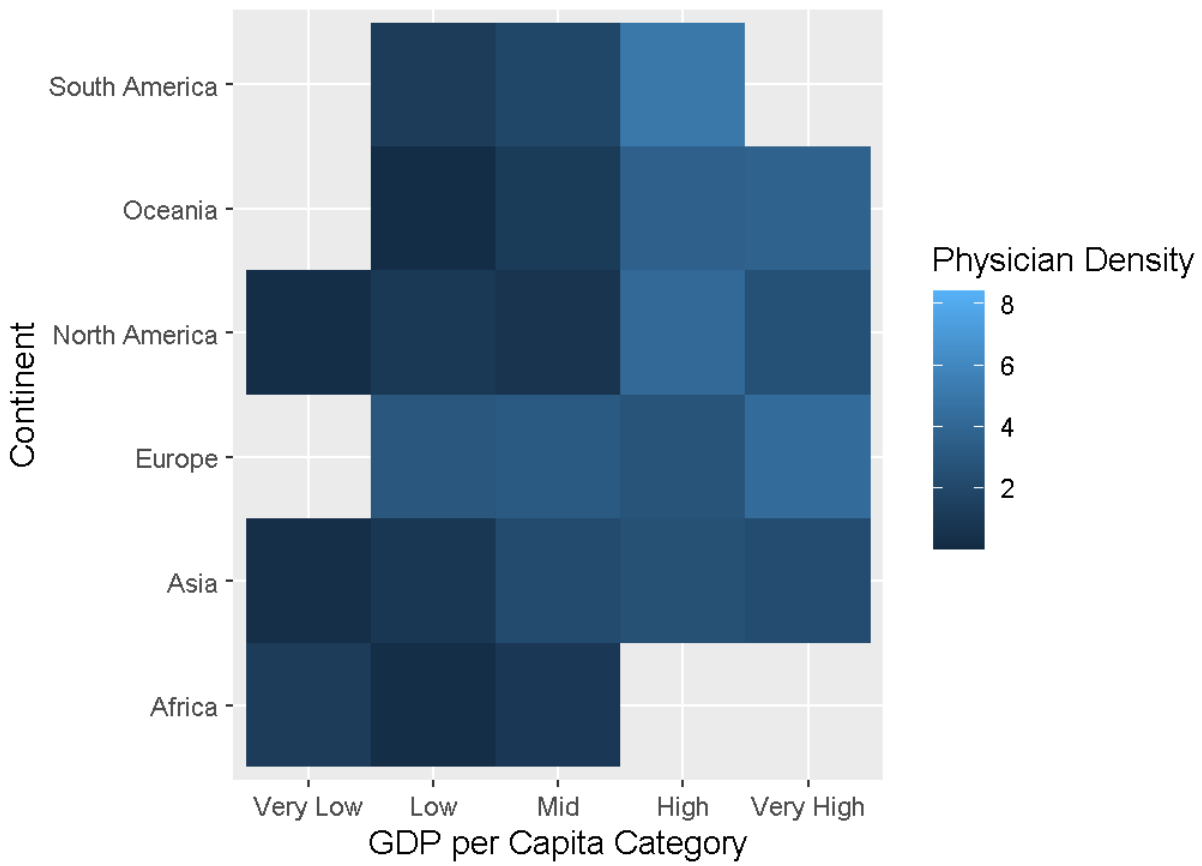
Africa	Asia	Europe	North America	Oceania	South America
52	47	43	23	14	12

HEATMAPS

Physician Density

Higher GDP per Capita tends to show a slightly higher physician density. However, some outliers may exist where high GDP does not correspond to high physician density. Lighter blue tiles appear mostly in the High and Very High GDP categories, suggesting wealthier countries tend to have more physicians per capita. For example, Europe and North America, categorized as High and Very High, show lighter shades, indicating better healthcare access. Countries with lower GDP per Capita usually have fewer physicians. The trend is especially clear in Africa and Asia, where lower financial resources often mean fewer healthcare resources. Most countries in the Very Low and Low GDP groups show this pattern, but there are a few exceptions. Some countries in these groups have more physicians than expected. This suggests that local factors can affect how many physicians are available, even when money is limited.

```
ggplot(data = edu_money, aes(x = Continent, y =GDP per Capita Category, fill =Physician Density)) +
geom_tile()
```

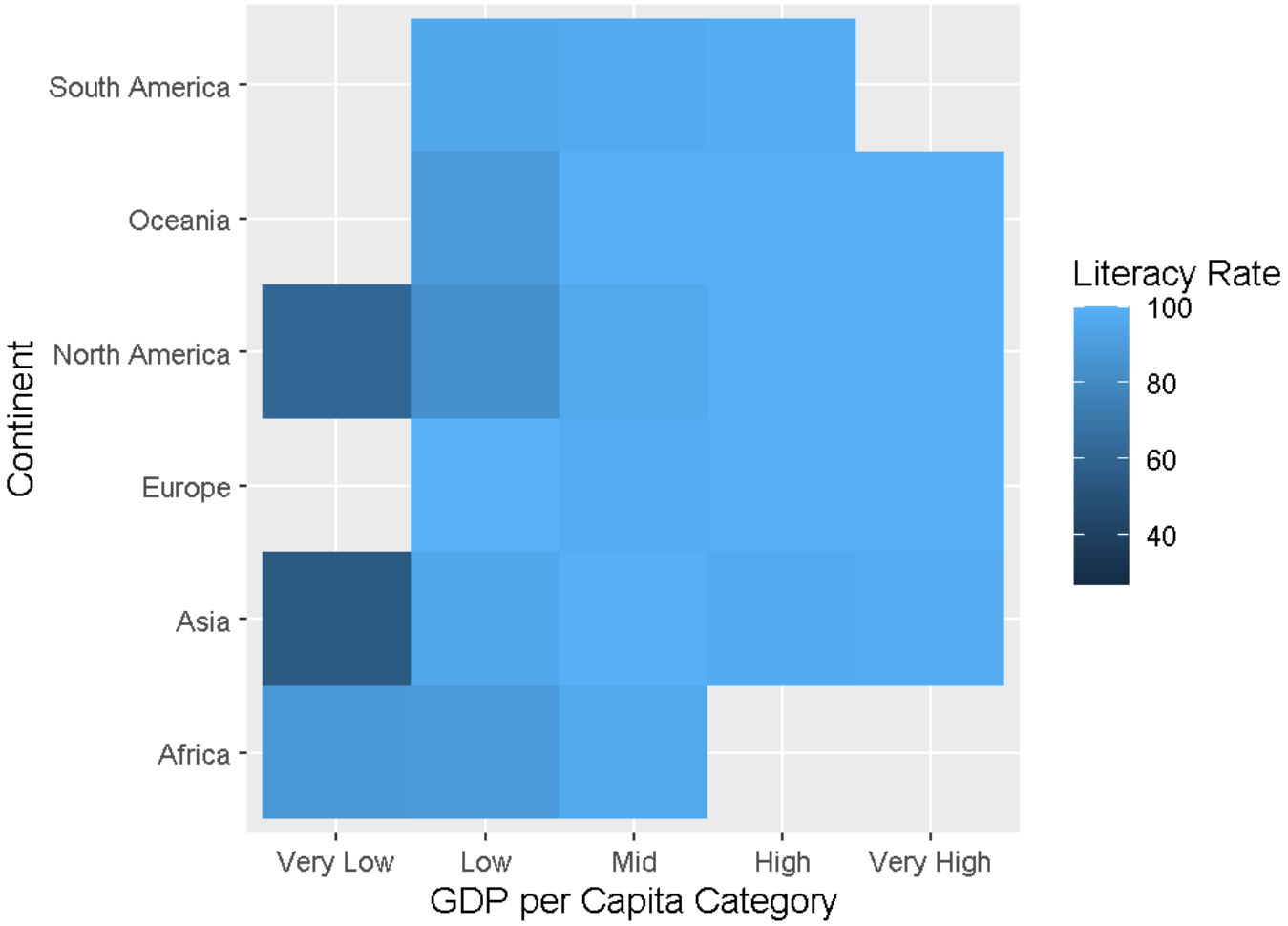


HEATMAPS

literacy rates

Higher GDP is showing higher literacy. Specifically, regions that fall in the High and Very High GDP categories show lighter shades, indicating literacy rates close to 100 percent. In contrast, low GDP are showing lower literacy. For example, Africa and Asia in the Very Low GDP categories have darker shades, suggesting literacy rates around 40–60 percent. Meanwhile, all countries in the Mid to High GDP categories show medium-light blue, meaning literacy rates are relatively high but not perfect.

```
> ggplot(data = edu_money, aes(x = `GDP per Capita Category`, y =Continent,
+                              fill =`Literacy Rate`)) + geom_tile()
```

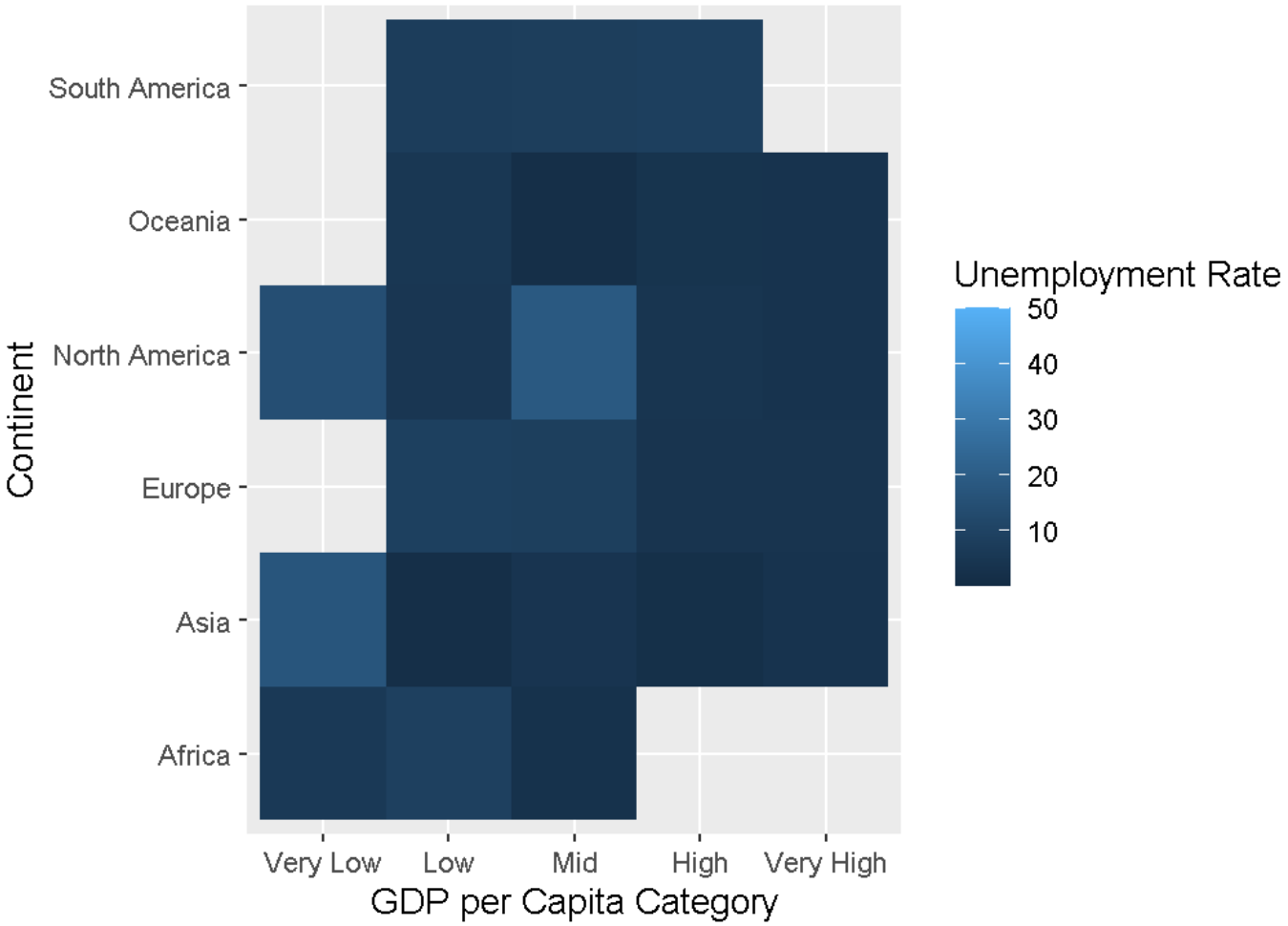


HEATMAPS

Unemployment

Darker shades indicate lower unemployment, while lighter shades indicate higher unemployment. The relationship is not showing a simple linear pattern. For instance, North America's Mid GDP regions have higher unemployment (lighter shade) compared to Africa's Very Low GDP regions, which display lower unemployment (darker shade). This highlights that some mid-GDP regions have higher unemployment than both very low and very high GDP regions.

```
> ggplot(data = edu_money, aes(x = `GDP per Capita Category`, y =Continent,
+                              fill =`Unemployment Rate`)) + geom_tile()
```

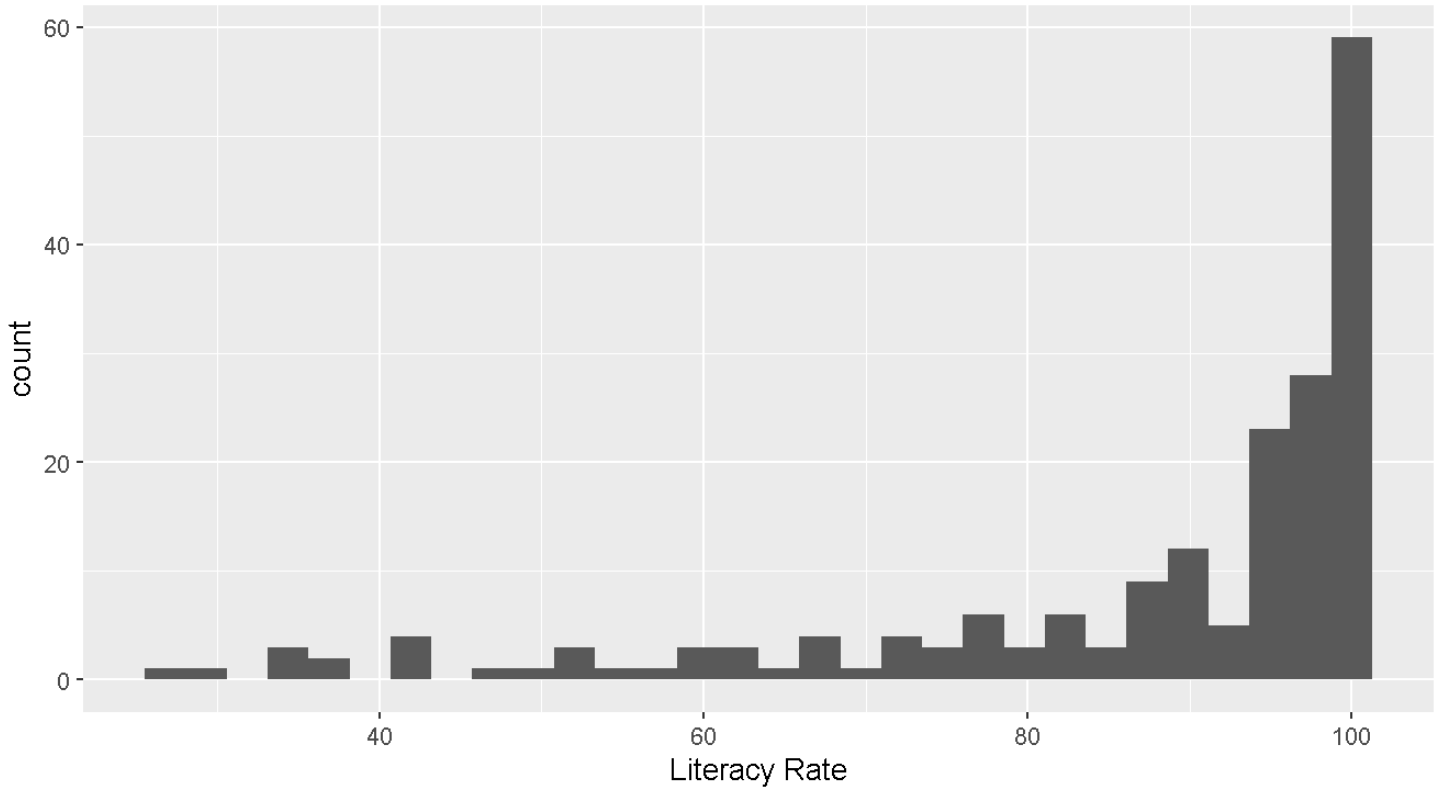


HISTOGRAMS

Literacy rate

The shape of the literacy rate histogram shows a Likely possibility for left skewness. It shows that Most countries have a good percentage of the population reading, but it slopes down to a marginalized group that has less than 50 percent of the population reading.

```
> ggplot(edu_money)+
+  geom_histogram(aes(x = `Literacy Rate`))
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

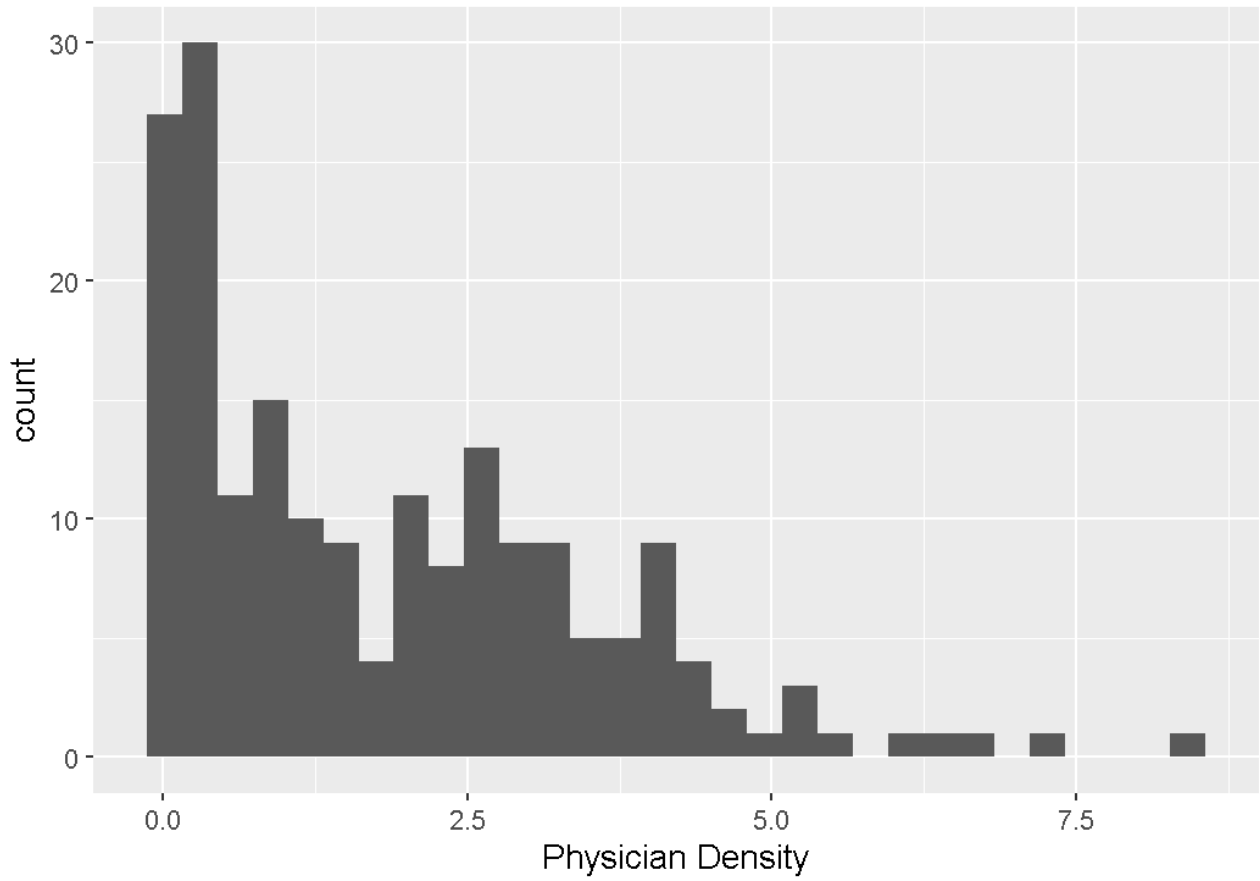


HISTOGRAMS

Physician Density Histogram

The shape of the Physician density histogram suggests right skewness. Most countries have low physician density, and a few countries appear to have much higher densities, possibly as outliers.

```
ggplot(edu_money)+
+  geom_histogram(aes(x = `Physician Density`))
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

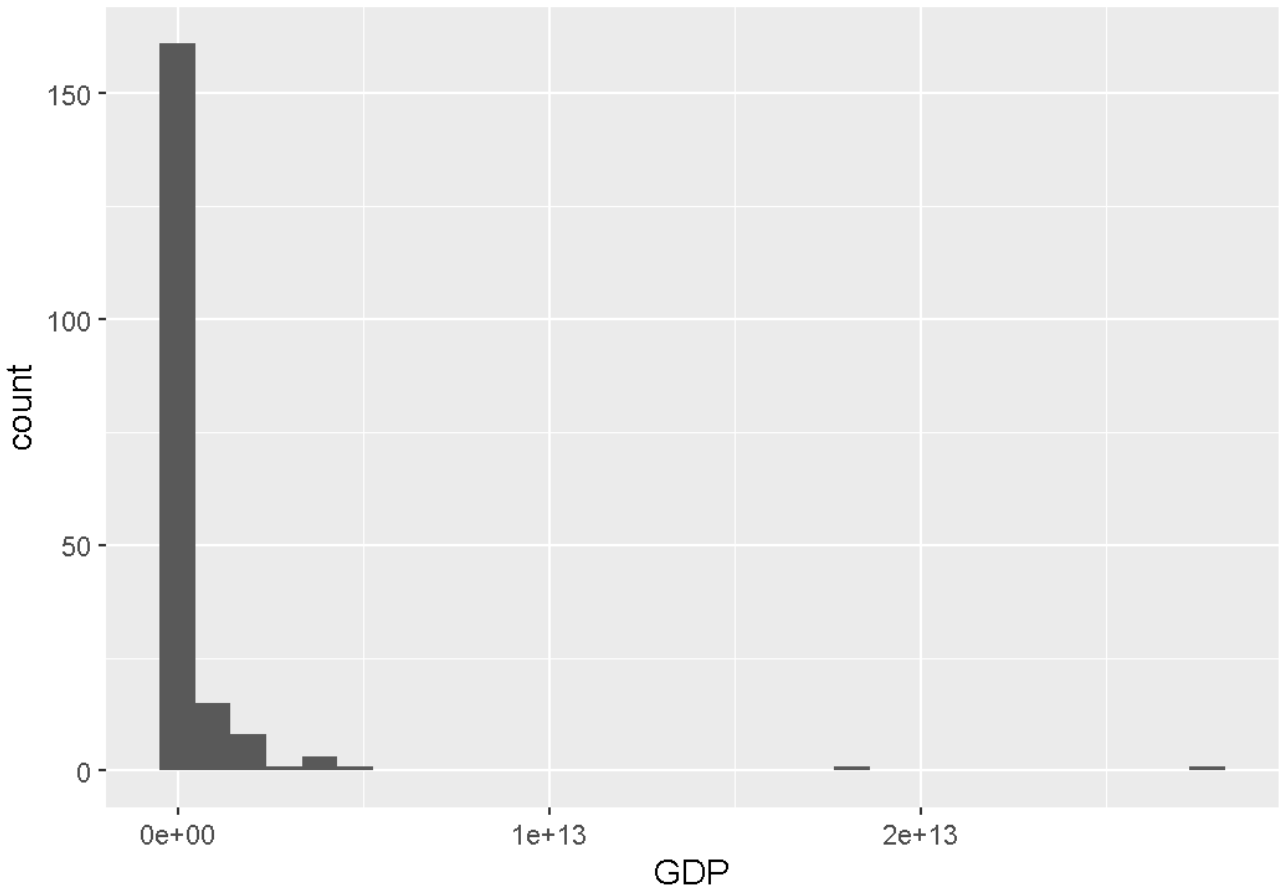


HISTOGRAMS

GDP Histogram

The GDP histogram looks like it has the least amount of variation, which may be due to the obvious outliers. This will definitely need to be further analyzed.

```
ggplot(edu_money)+
+  geom_histogram(aes(x = GDP))
`stat_bin()` using `bins = 30`. Pick better value `binwidth`
```



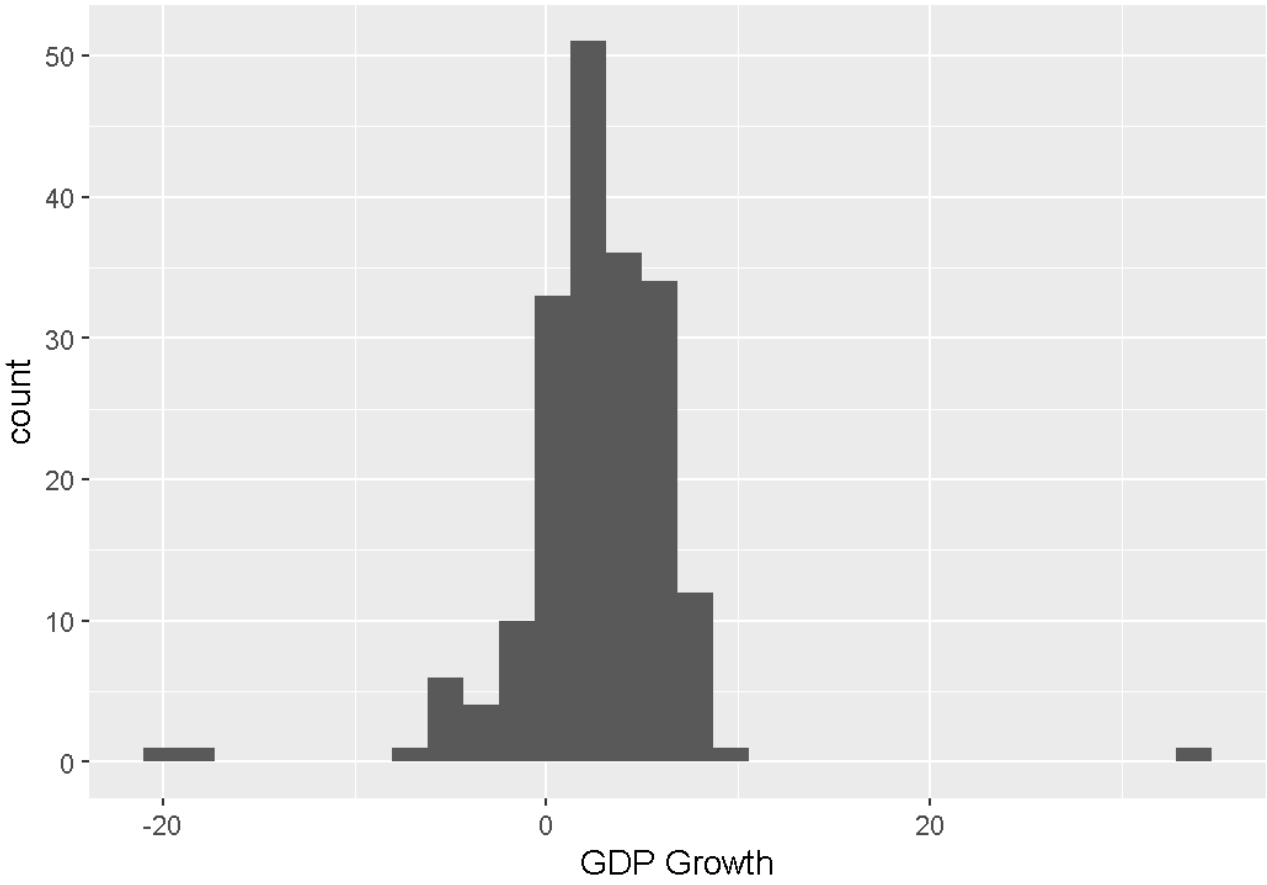
.

HISTOGRAMS

GDP Growth Histogram

The GDP growth histogram is the most symmetric among the datasets. The data demonstrates a clear central tendency, with most countries experiencing minimal changes in GDP growth. However, there are notable outliers, with some countries experiencing significant increases or decreases in growth.

```
ggplot(edu_money)+
+  geom_histogram(aes(x =`GDP Growth`))
`stat_bin()` using `bins = 30`. Pick better value `binwidth`
```

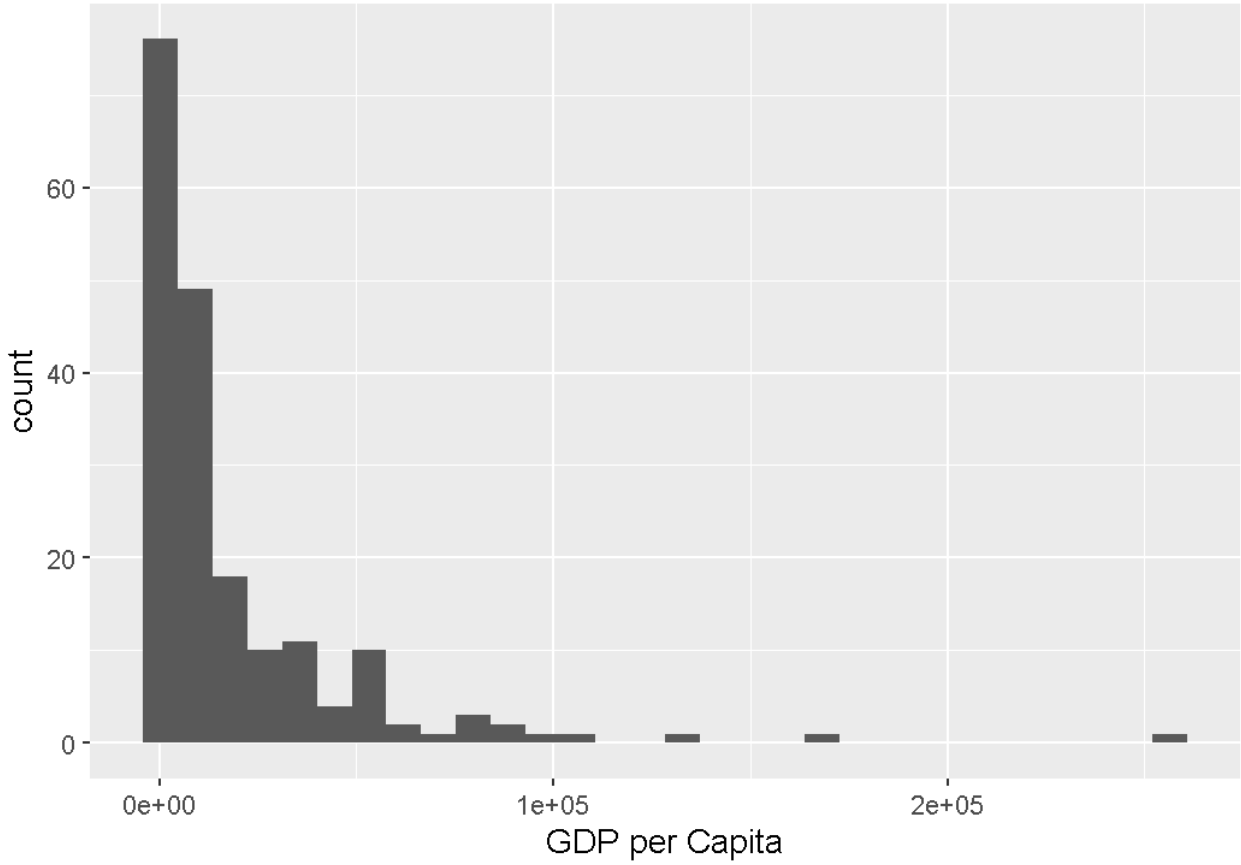


HISTOGRAMS

GDP per Capita Histogram

This histogram displays a right-skewed distribution, indicating that most countries have low GDP per capita. It also reveals the presence of outliers, with a few countries exhibiting a very high GDP per capita.

```
> ggplot(edu_money)+
+  geom_histogram(aes(x =`GDP per Capita`))
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

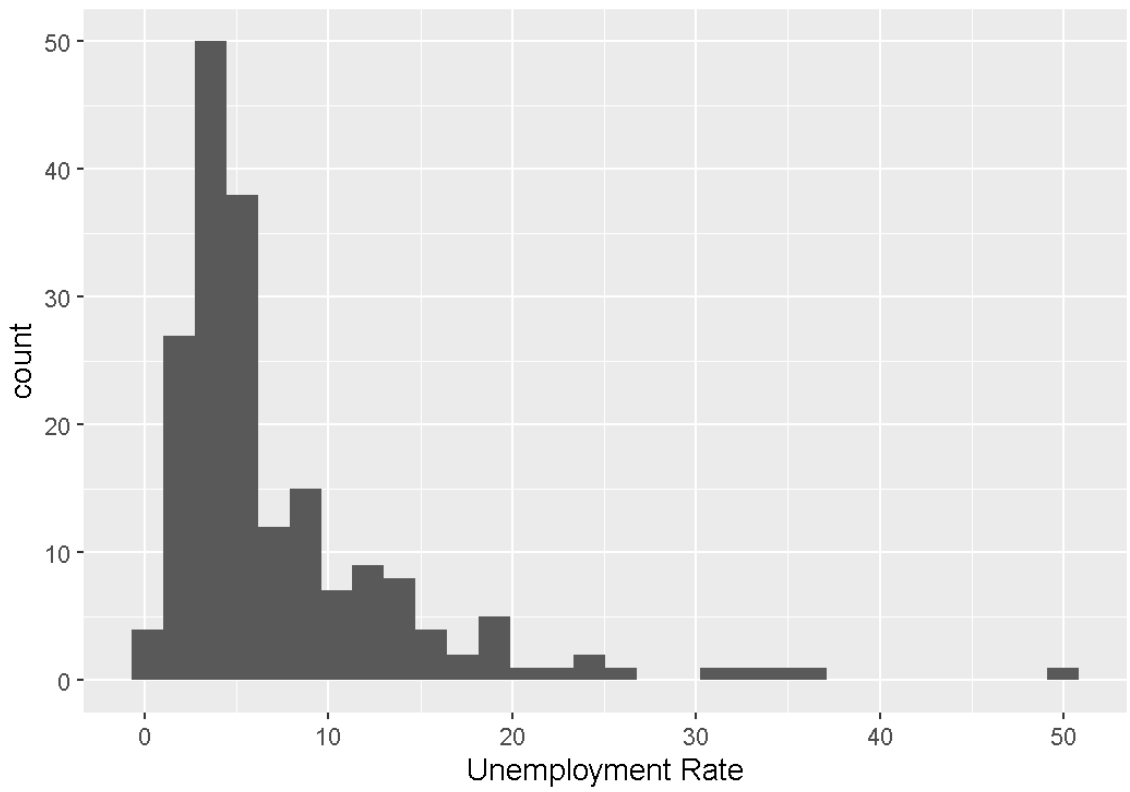


HISTOGRAMS

Unemployment Rate Histogram

This histogram shows the right skewness, and it's easy to see that most countries have low unemployment rates, with a few outliers having around 50 percent.

```
> ggplot(edu_money)+
+  geom_histogram(aes(x = `Unemployment Rate`))
`stat_bin()` using `bins = 30`. Pick better value `binwidth
```

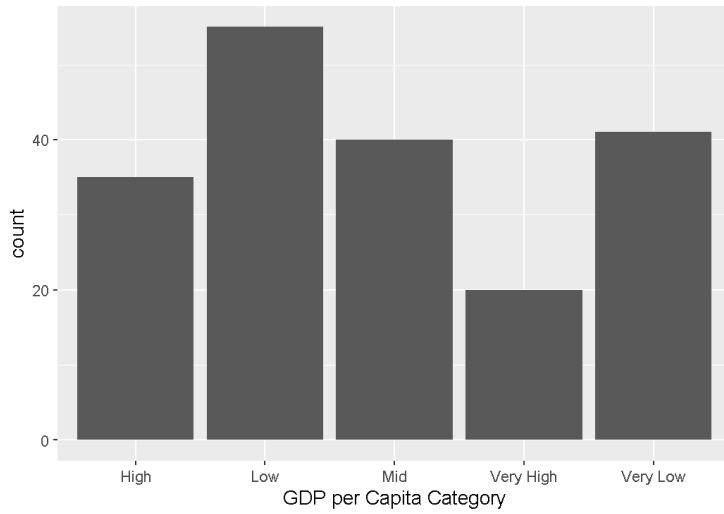


BAR PLOTS

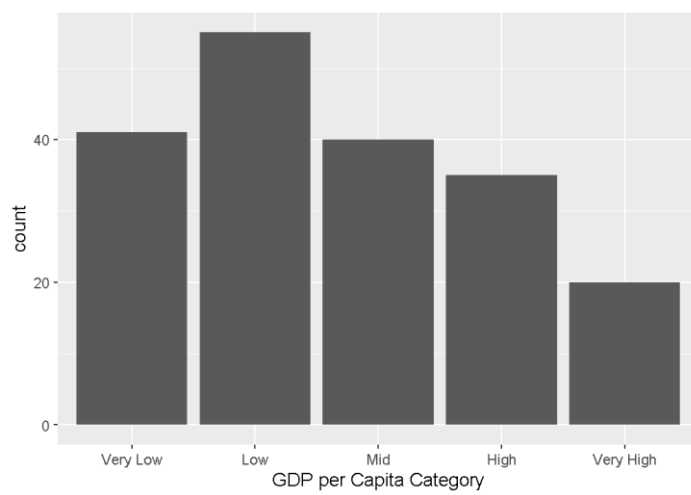
GDP per Capita Category

The GDP per capita is broken down into 4 categories of very low, low, mid, high, and very high. I ordered the data from very low to very high to clearly display its distribution, highlighting that most cases fall into the low and very low categories. The very high category contains the fewest data points.

```
> ggplot(edu_money)+
+  geom_bar(aes(x =`GDP per Capita Category`))
```




```
> education_economy_data <- read_csv("education/education-economy-data.csv",
+    col_types = cols(`GDP per Capita Category` = col_factor(levels = c("Very Low",
+        "Low", "Mid", "High", "Very High"))))
```

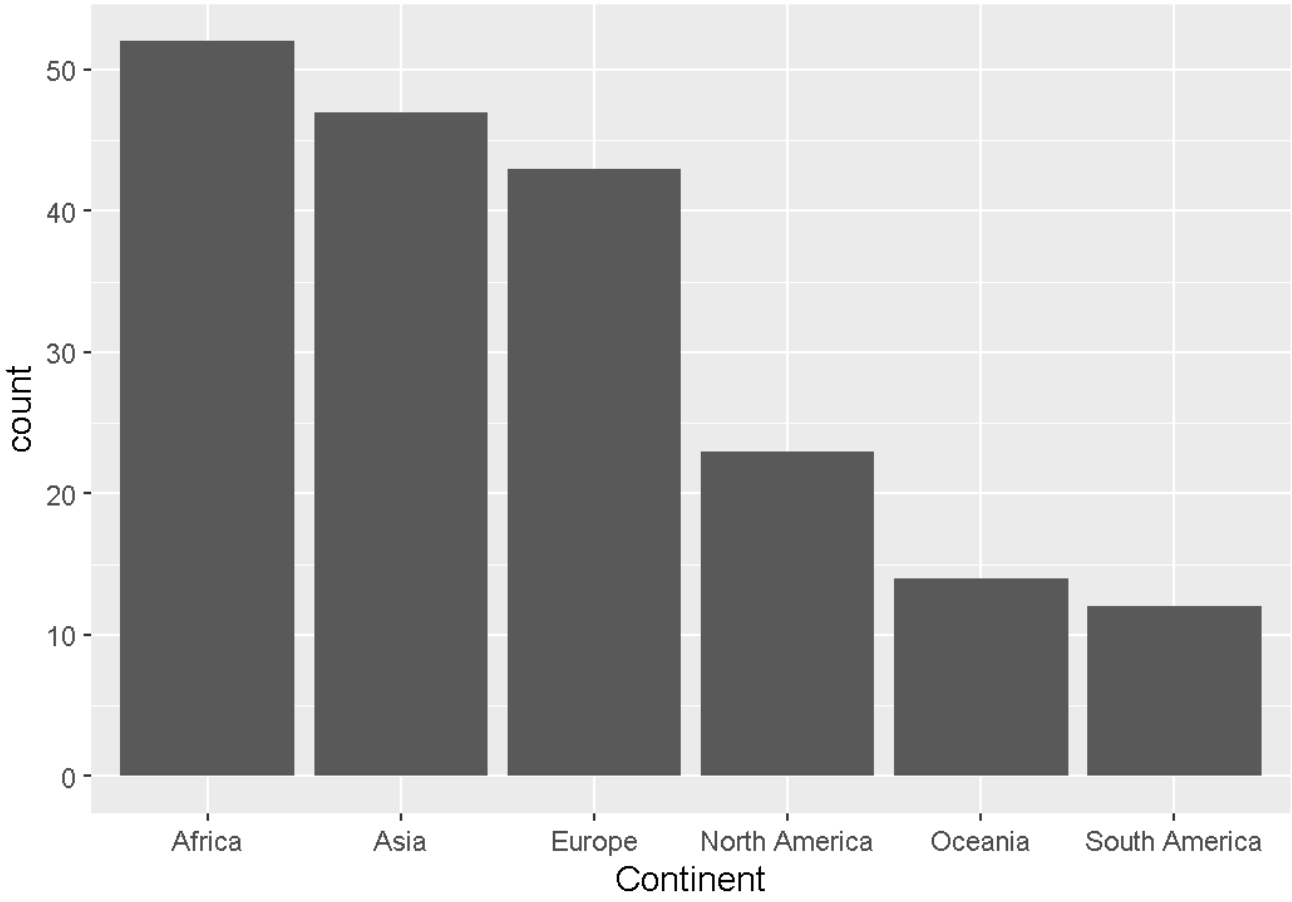


BAR PLOTS

Continents

This bar graph shows the number of variables measured for each continent. According to the chart, Africa, Asia, and Europe account for the majority of the data. (Continents of the World by Population, Area & Density (2025 Data), 2023) Note that some regions represent small geographic areas; for example, Oceania includes Australia and several other islands. Also, this bar graph does not represent population size.

```
> ggplot(edu_money)+
+  geom_bar(aes(x = Continent))
```



SUMMARY OF STATISTICS

In this table, I ran three separate codes and then combined and configured the data into one table to get the most useful information. Most variables are showing a likeliness for right skewness (meaning the majority of values are clustered toward the lower end, with a few much larger values), such as physician density, GDP, GDP per capita, and unemployment rate. The biggest variation (or spread in values) is seen in GDP, while GDP growth has the lowest difference between mean and median values. The literacy rate is the only variable showing skewness to the left, indicating that most values are high with a few lower outliers.

```
> sapply(edu_money[, sapply(edu_money, is.numeric)], summary)
> sapply(edu_money[, sapply(edu_money, is.numeric)], sd)
install.packages("EnvStats")
library(EnvStats)
> sapply(edu_money[, sapply(edu_money, is.numeric)], cv)
```

	Literacy Rate	Physician Density	GPD	GPD Growth	GPD per capita	Unemployment Rate
Minimum	26.8000	0.010000	6.228031e+07	-20.111576	193.0071	0.130000
1st quartile	81.8000	0.345000	1.356848e+10	0.979868	2284.2236	3.139500
Median	95.6000	1.470000	4.260088e+10	2.733798	6947.3593	5.264000
Mean	86.9947	1.848639	5.427605e+11	2.666018	18796.7951	7.522831
3erd Quartile	99.0000	2.950000	2.651225e+11	4.987486	22322.3550	9.027000
Maximum	100.000	8.420000	2.772070e+13	33.802617	256580.515	50.000000
Standard deviation	1.762210 e+01	1.679639 e+00	2.452696 e+12	4.362681 e+00	3.053529 e+00	7.092726 e+00
Coefficient of variation	0.202565	0.9085818	4.5189282	1.6364032	1.6244943	0.9428268

To break down the table a little more, this is the focus points from each category:

Literacy Rate

- The literacy rate is very stable across countries, with a coefficient of variation of approximately 0.20.
- Most nations have high literacy rates, with a median of 95.6%, but there are some countries with extremely low literacy rates, the minimum being 26.8%.

Physician Density

- The Physician density shows moderate variability, with a coefficient of variation of approximately 0.91.
- The median physician density is 1.47 physicians per 1,000 people, but with a maximum of 8.42, this indicates that some countries have much higher access to
  Healthcare.

GDP

- The GDP is extremely variable across countries, with a coefficient of variation of approximately 4.52 compared to the mean of 5.427605e+11.
- The maximum GDP of $27.72 trillion is far beyond the mean of $542.76 billion, indicating strong outliers.

GDP Growth

- The GDP growth shows moderate variability compared to the mean, with a coefficient of variation of approximately 1.64.
- A negative minimum GDP growth of -20.11% suggests severe economic contraction in some countries, while a maximum of 33.8% demonstrates rapid growth in others.

GDP per Capita

- GDP per capita shows high variability compared to the mean, with a coefficient of variation of approximately 1.62.
- The maximum GDP per capita of $256,580 compared to the median of $6,947 highlights extreme inequality between wealthy and poorer nations.

Unemployment Rate

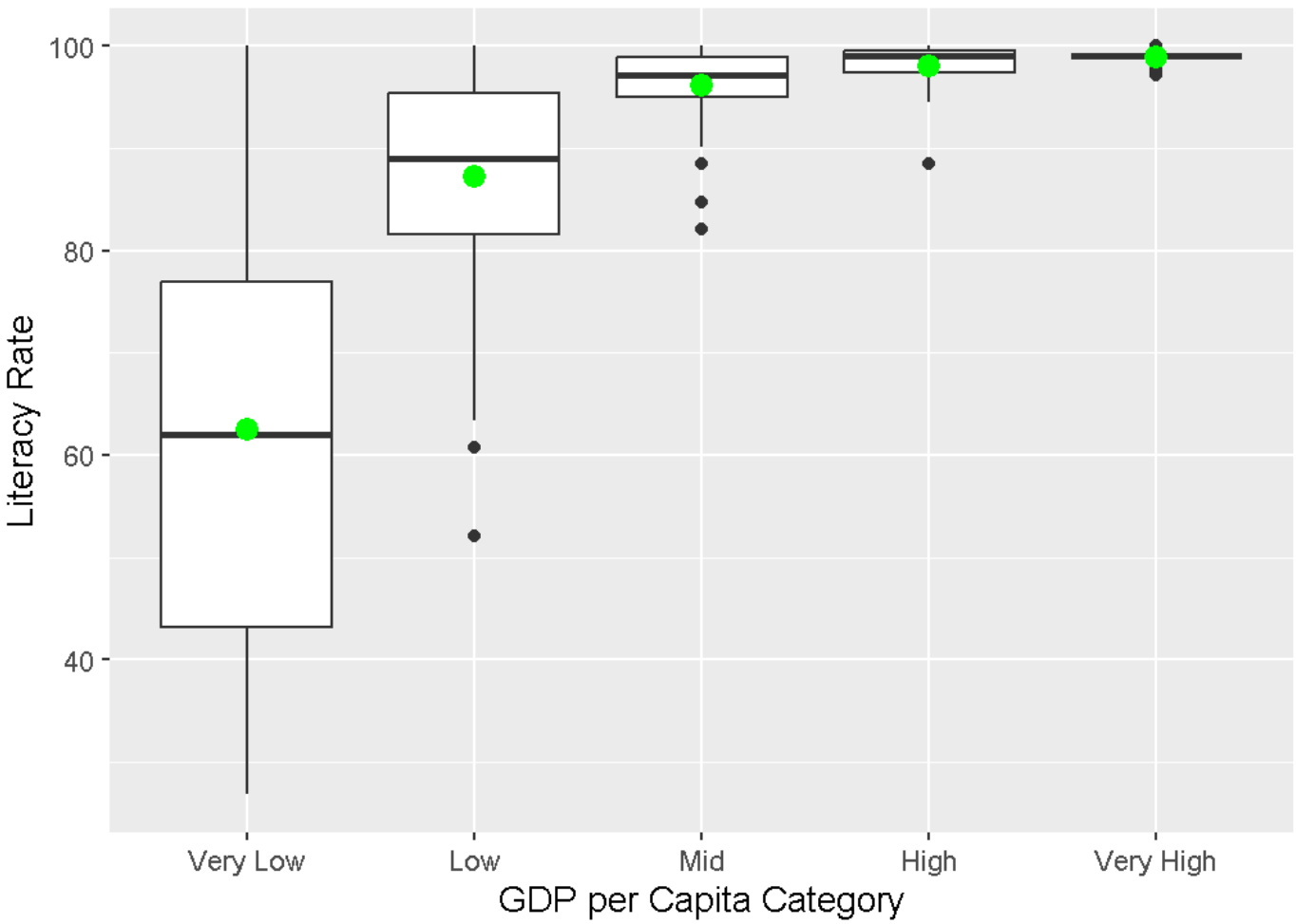
- The unemployment rate has moderate variability, with a coefficient of variation of approximately 0.94.
- A maximum unemployment rate of 50% is a major outlier compared to the median of 5.26%.

OUTLIERS/BOX PLOTS

GDP per Capita Category x Literacy Rate

The General Trend is that as GDP per capita rises, literacy rates increase. Countries with Very High GDP per capita have literacy rates close to 100%, while those with Very Low GDP per capita show an average of much lower and more variable literacy rates. The Median literacy rate rises steadily from left to right. The green dots represent mean values for each category. They confirm the upward trend: mean literacy rate increases with GDP. The low category has the most skewness.

```
ggplot(edu_money)+
+  geom_boxplot(aes(x = `GDP per Capita Category`, y = `Literacy Rate`))+
+  stat_summary(aes(x = `GDP per Capita Category`, y = `Literacy Rate`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

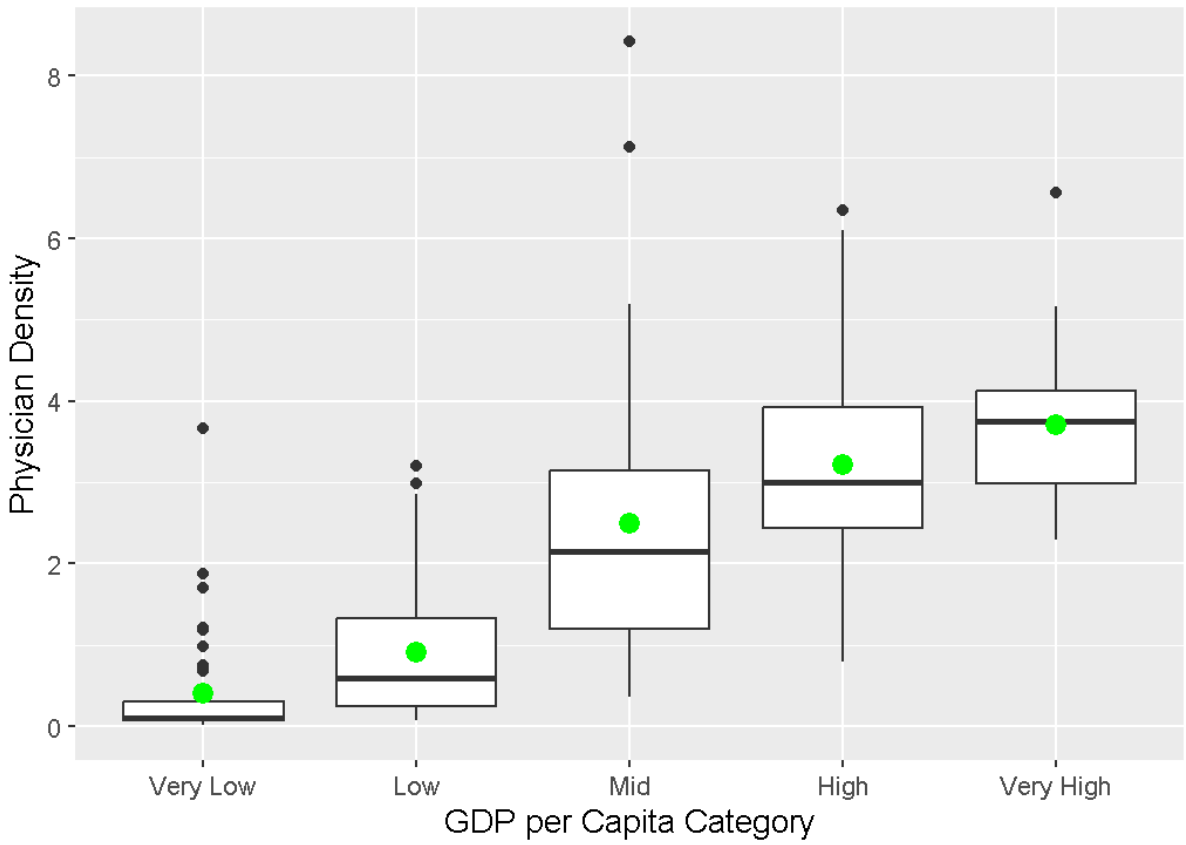


OUTLIERS/BOX PLOTS

GDP per Capita Category x Physician Density

The trend of this boxplot is that as the GDP per capita category rises, the Physician density increases. Variability grows with GDP. Very Low and Low categories have small ranges. The mid and High categories show a wider spread and more outliers. Some countries in the Mid and High GDP categories have unusually high physician density. The very low category shows the most outliers and skewness, while the very high GPD shows the least amount of outliers and skewness.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = `GDP per Capita Category`, y = `Physician Density`))+
+  stat_summary(aes(x = `GDP per Capita Category`, y = `Physician Density`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

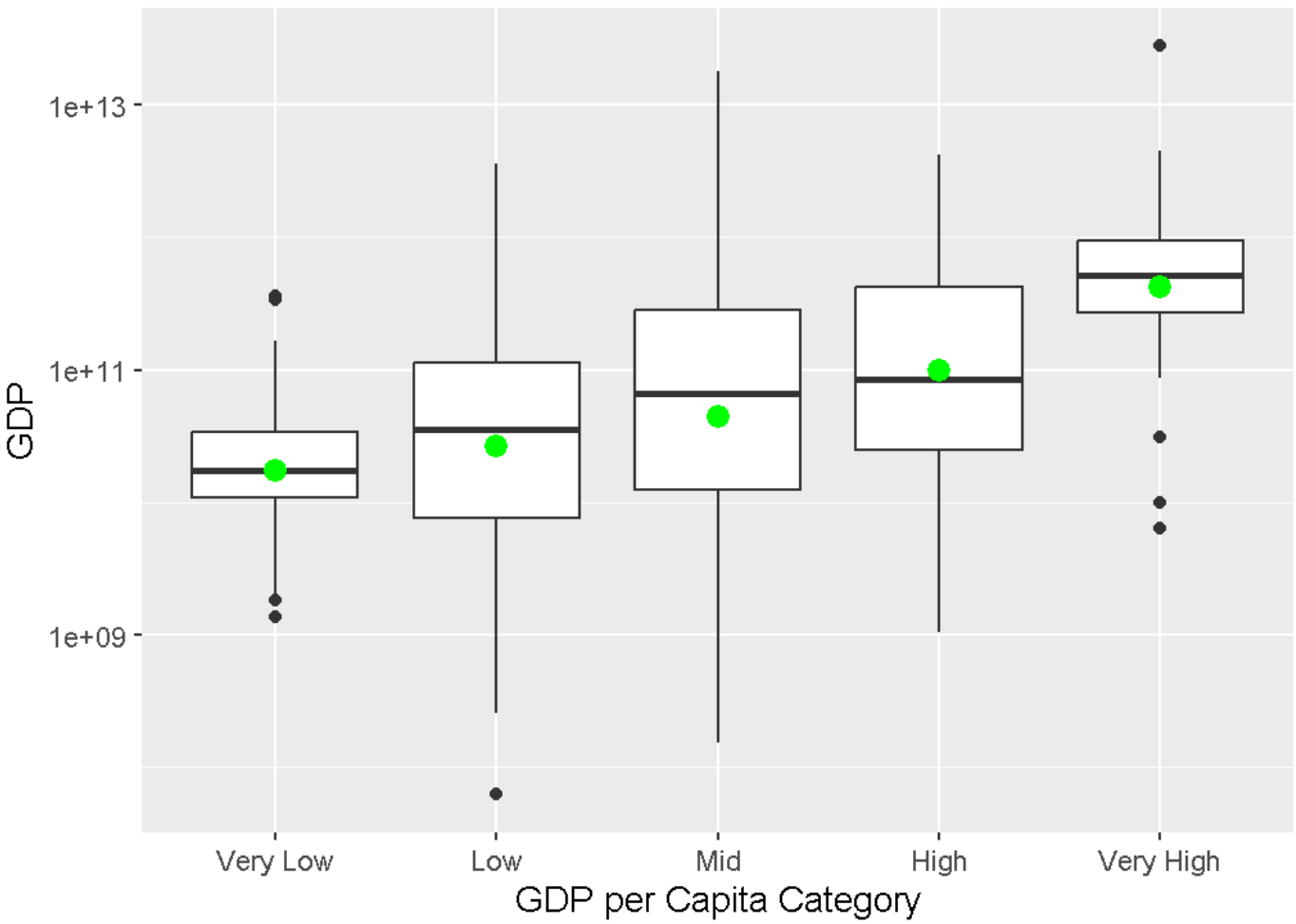


OUTLIERS/BOX PLOTS

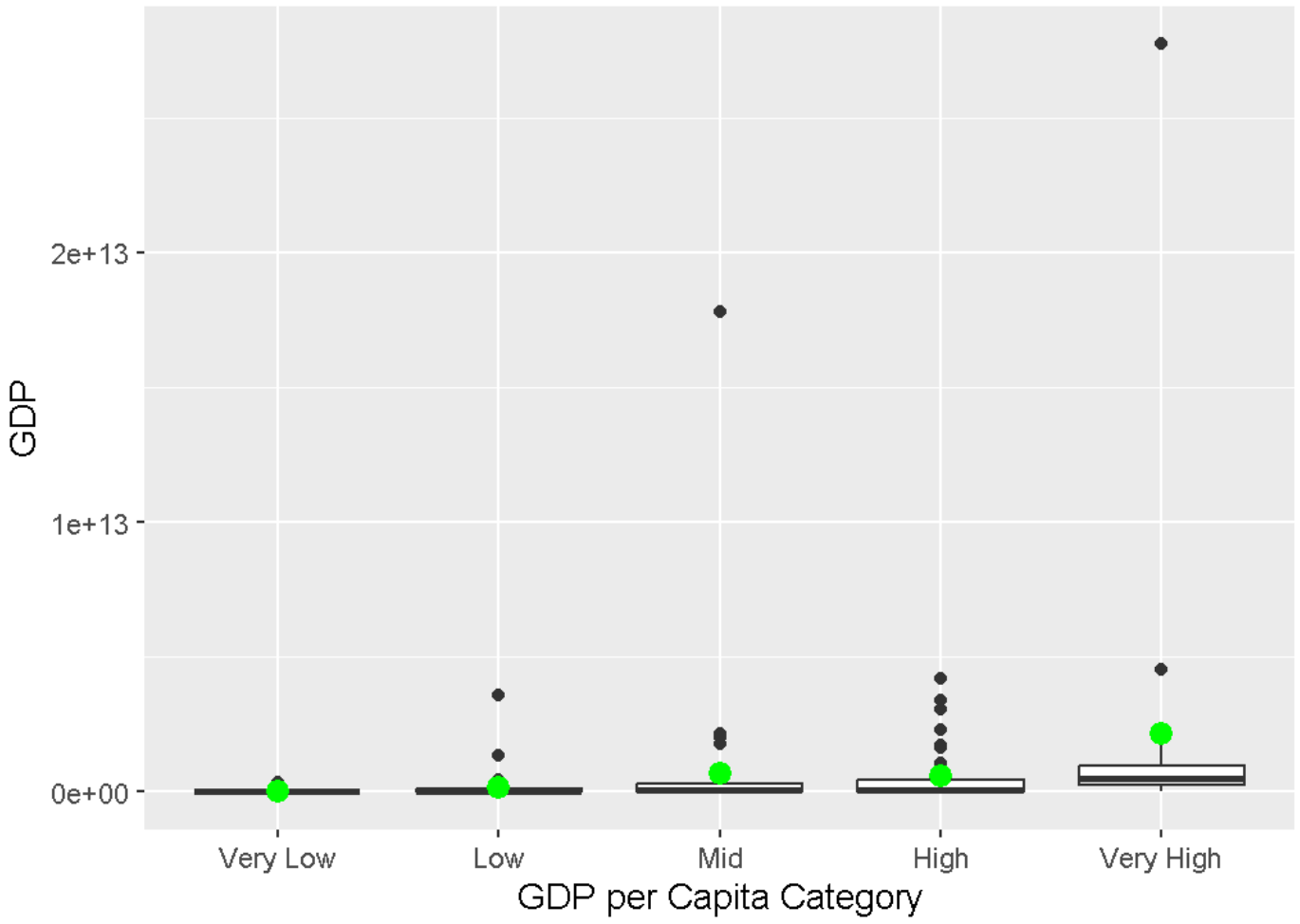
GDP per Capita Category x GPD

Countries with higher GDP per capita tend to have higher total GDP, but the relationship is not linear, because population size and economic structure matter. The log scale makes the pattern clearer and reduces distortion from extreme outliers, but it does change the mean of the data, especially the mid, high, and very high categories.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = `GDP per Capita Category`, y = GDP))+
+  scale_y_log10()+
+  stat_summary(aes(x = `GDP per Capita Category`, y = GDP), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```




```
ggplot(edu_money)+
+  geom_boxplot(aes(x = `GDP per Capita Category`, y = GDP))+
+  stat_summary(aes(x = `GDP per Capita Category`, y = GDP), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

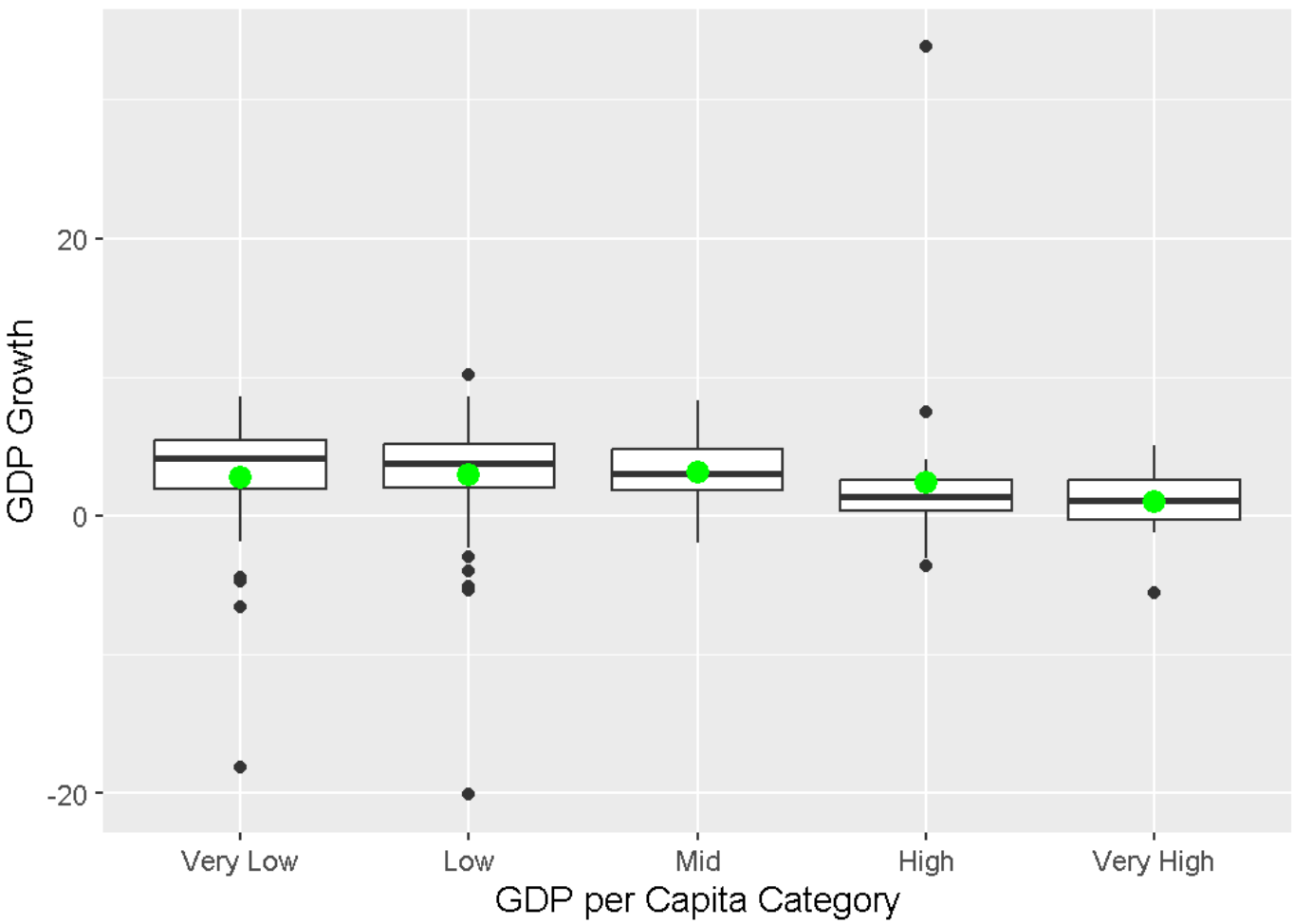

2e+13
1e+13
0e+00
GDP
Very Low
Low
Mid
High
Very High
GDP per Capita Category

OUTLIERS/BOX PLOTS

GDP per Capita Category x GDP Growth

Median GDP Growth is positive for all categories, but slightly decreases as GDP per capita increases. Very Low, Low, and Mid categories have similar medians around 3–4 percent growth. The High and Very High categories show lower median growth around 2–3 percent. Very Low and Low categories have some extreme negative outliers around -20 percent. The high category has a very large positive outlier above 30 percent. Very Low and Low categories have wider spreads, indicating more variability. Very High category is more compact, suggesting stable growth.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = `GDP per Capita Category`, y = `GDP Growth`))+
+  stat_summary(aes(x = `GDP per Capita Category`, y = `GDP Growth`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

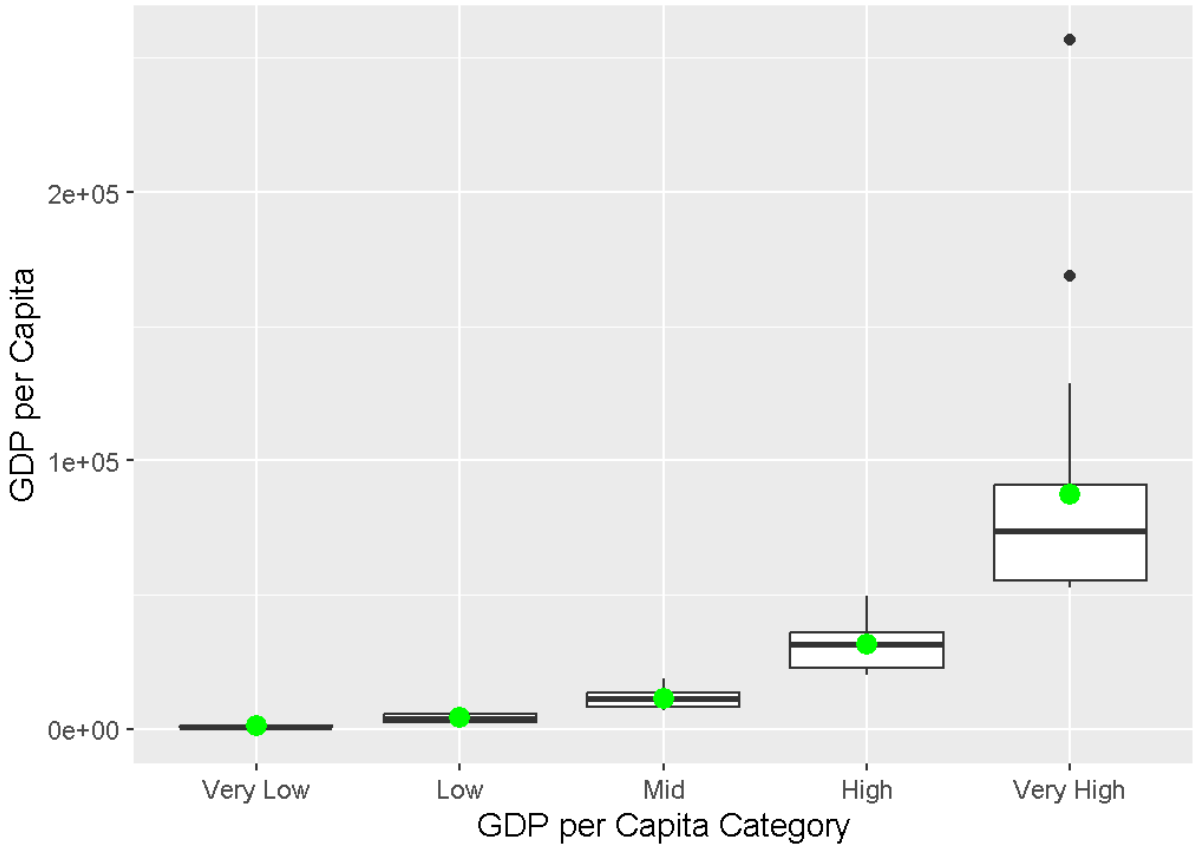


OUTLIERS/BOX PLOTS

GDP per Capita Category x Unemployment Rate

The Very Low and Low categories have GDP per capita values near zero, but they do show some variation. The Mid category shows a slight increase, but values remain relatively low. The High category demonstrates a noticeable jump in GDP per capita. In the Very High category, there is a huge spread, with some extreme outliers pulling the mean far above the median. GDP per capita grows exponentially across categories. This graph was generated to ensure all countries were categorized correctly.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = `GDP per Capita Category`, y =`GDP per Capita`))+
+  stat_summary(aes(x = `GDP per Capita Category`, y =`GDP per Capita`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

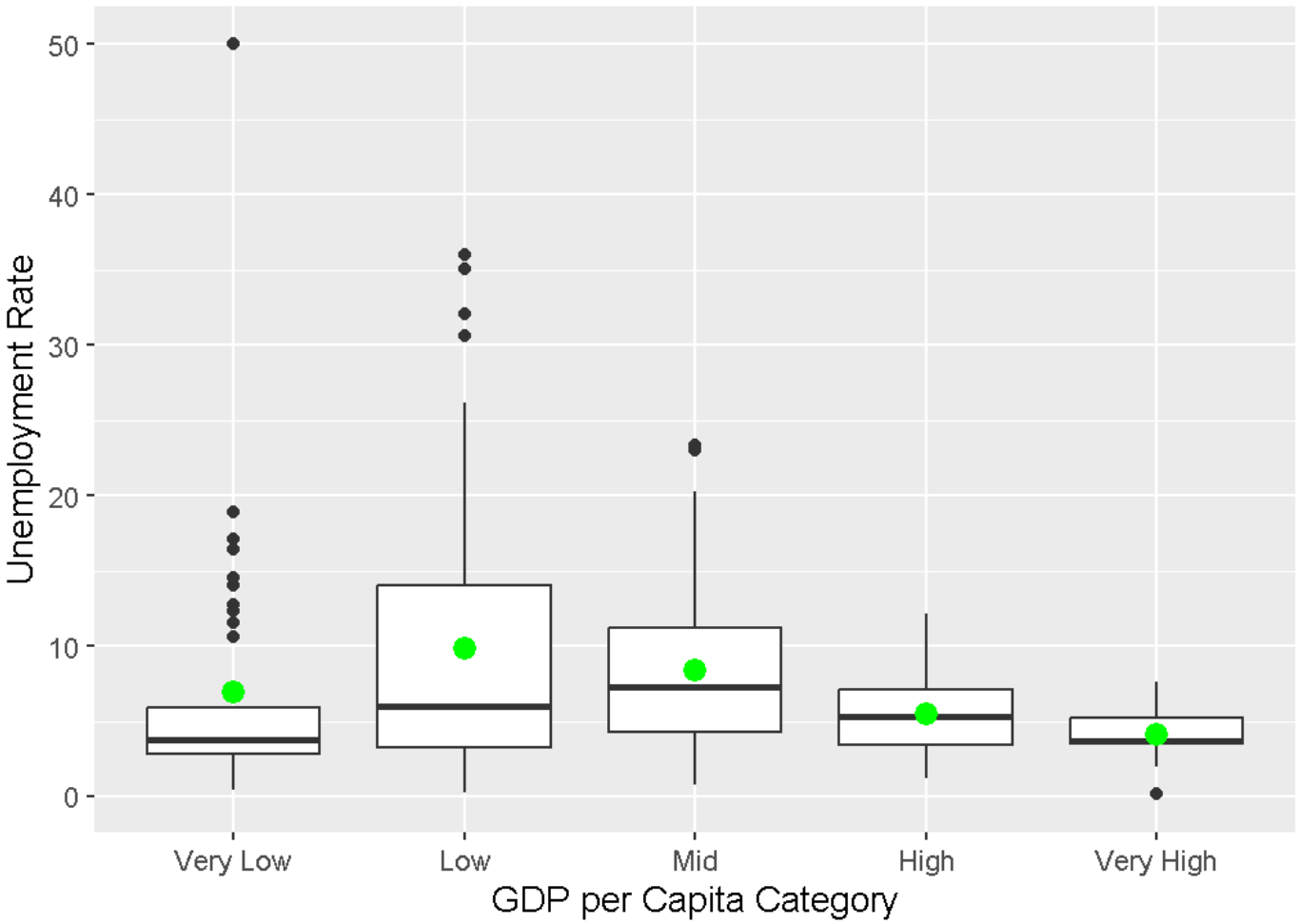


OUTLIERS/BOX PLOTS

GDP per Capita Category x Unemployment Rate

Countries with higher GDP per capita are showing a slightly lower unemployment rate and much less variability. The Very Low and Low GDP categories have wide ranges and numerous outliers. In contrast, the High and Very High GDP groups have tighter distributions, with median unemployment around 4–6 percent, suggesting more stable labor markets. The trend indicates a negative relationship between GDP per capita and the unemployment rate, with economic prosperity typically associated with lower, more consistent unemployment levels.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = `GDP per Capita Category`, y = `Unemployment Rate`))+
+  stat_summary(aes(x = `GDP per Capita Category`, y = `Unemployment Rate`),
+  fun= mean, geom = 'point', color = 'green', size = 3)
```

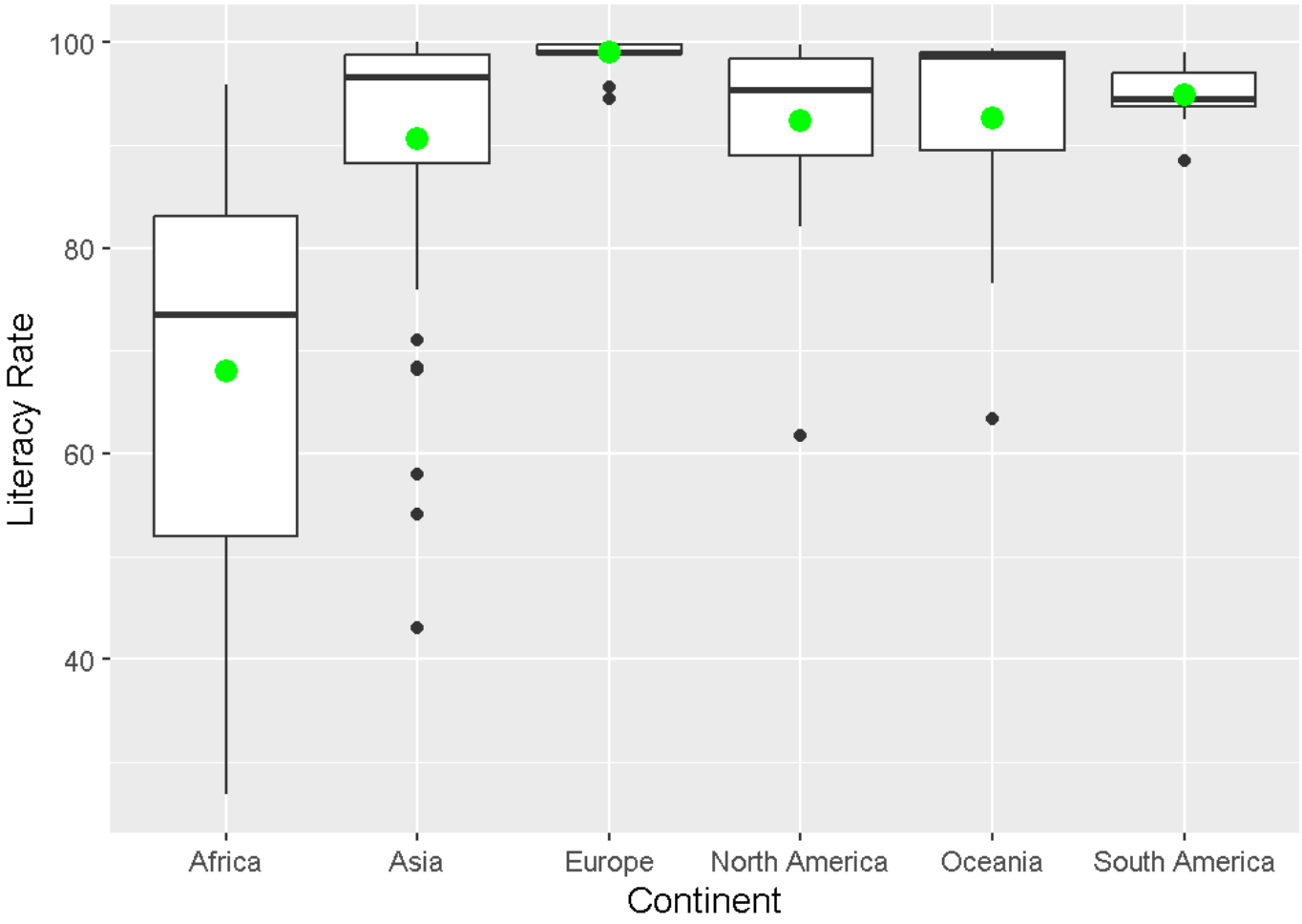


OUTLIERS/BOX PLOTS

Continents x Literacy Rate

Europe has the highest and most consistent literacy rates. Africa shows the greatest inequality in literacy rates. Other continents include Asia, North America, Oceania, South America. Generally, all of these have high literacy rates with occasional low outliers.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = Continent, y = `Literacy Rate`))+
+  stat_summary(aes(x = Continent, y = `Literacy Rate`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

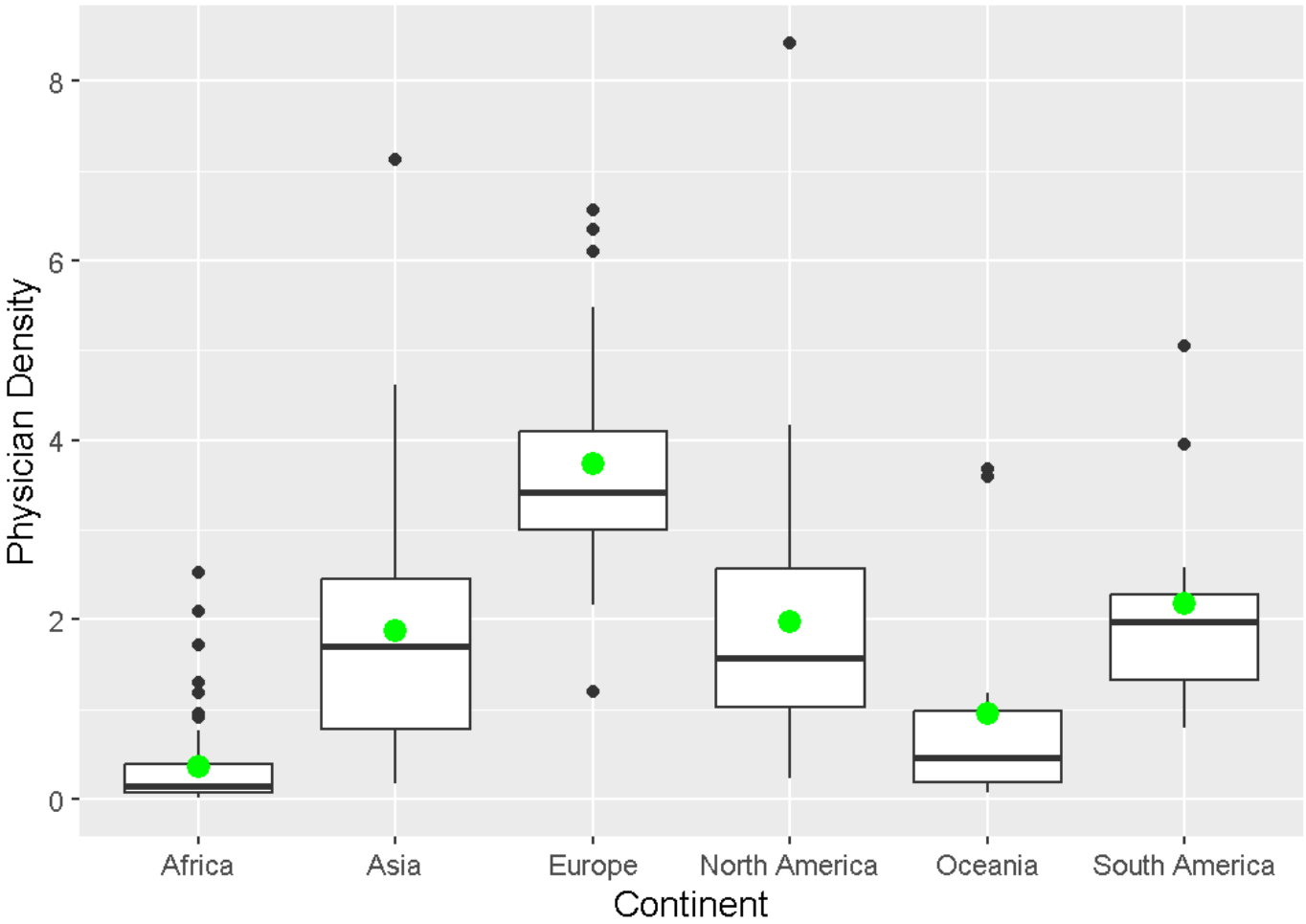


OUTLIERS/BOX PLOTS

Continents x Physician Density

Europe has the highest physician density overall. Africa has the lowest physician density. Oceania has low physician density, similar to Africa. There is a clear gradient from Africa (lowest) to Europe (highest).Asia and North America have moderate physician densities, but North America shows a wider spread and some very high outliers. South America is slightly higher than Asia. Variability is greatest in Asia and North America, suggesting uneven distribution within those continents. Outliers are present in all continents, but North America and Asia have extremely high outliers.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = Continent, y = `Physician Density`))+
+  stat_summary(aes(x = Continent, y = `Physician Density`), fun= mean,
+  geom = 'point', color = 'green', size = 3)
```

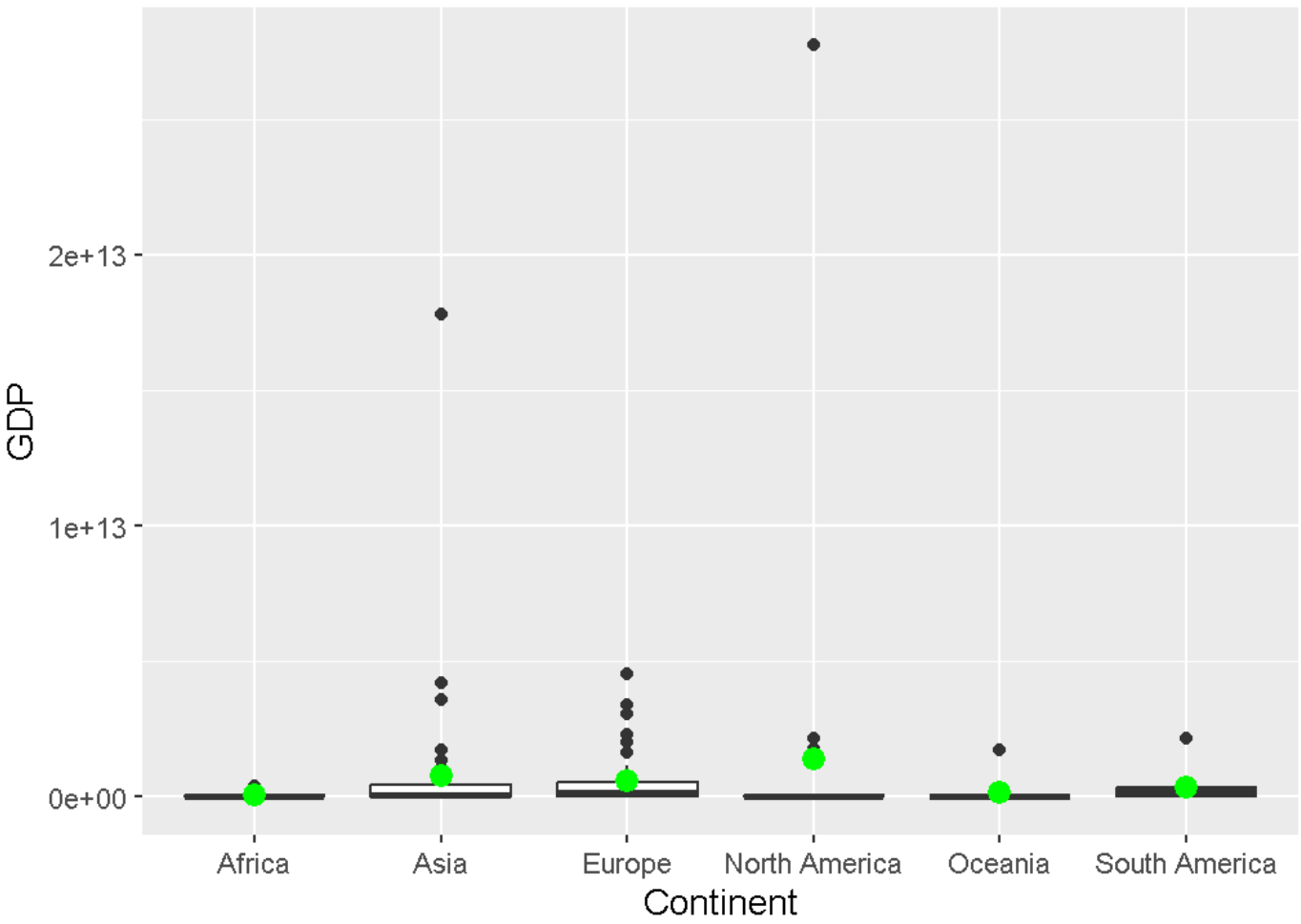


OUTLIERS/BOX PLOTS

Continents x GDP

Most GDP values are clustered near zero for all continents, as shown by the boxplots being very close to the bottom of the graph. There are many outliers far above the main cluster, indicating that a few countries have extremely high GDP compared to the majority. Highly skewed distribution. A few countries dominate global GDP, while most have much smaller values. North America, Asia, and Europe have the largest outliers. Africa and Oceania have the smallest GDP values overall.

```
geom_boxplot(aes(x = Continent, y = GDP))+
+  stat_summary(aes(x = Continent, y = GDP), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

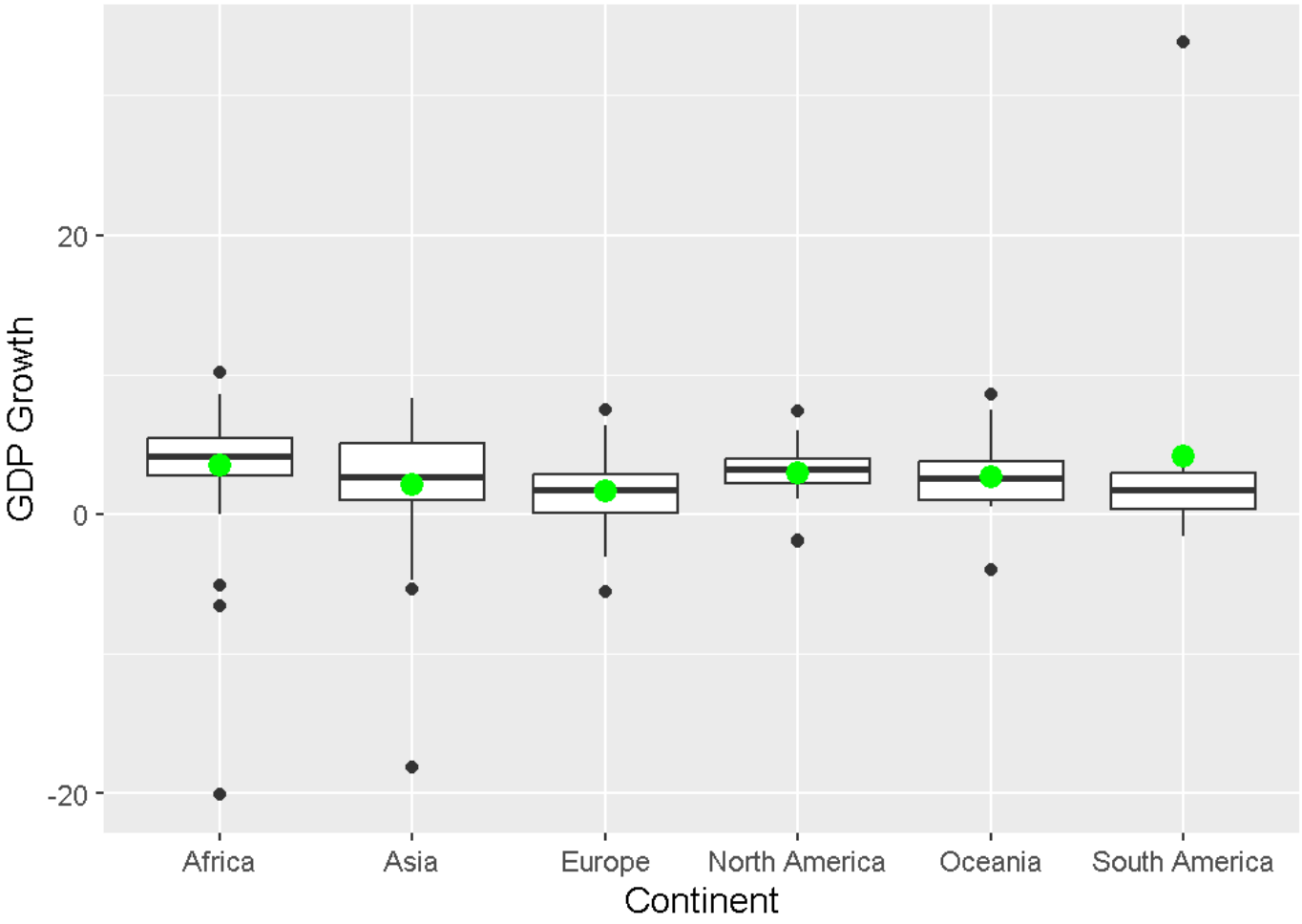


OUTLIERS/BOX PLOTS

Continents x GDP growth

Most continents have median and mean GDP growth slightly above zero, suggesting a small positive average growth rate. Asia and Africa show wider boxes and whiskers, indicating higher variability in GDP growth. Europe and South America have relatively narrower boxes, suggesting more consistent growth rates. Africa and Asia have negative outliers, while South America has one very high positive outlier, which is unusual compared to the others.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = Continent, y = `GDP Growth`))+
+  stat_summary(aes(x = Continent, y = `GDP Growth`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

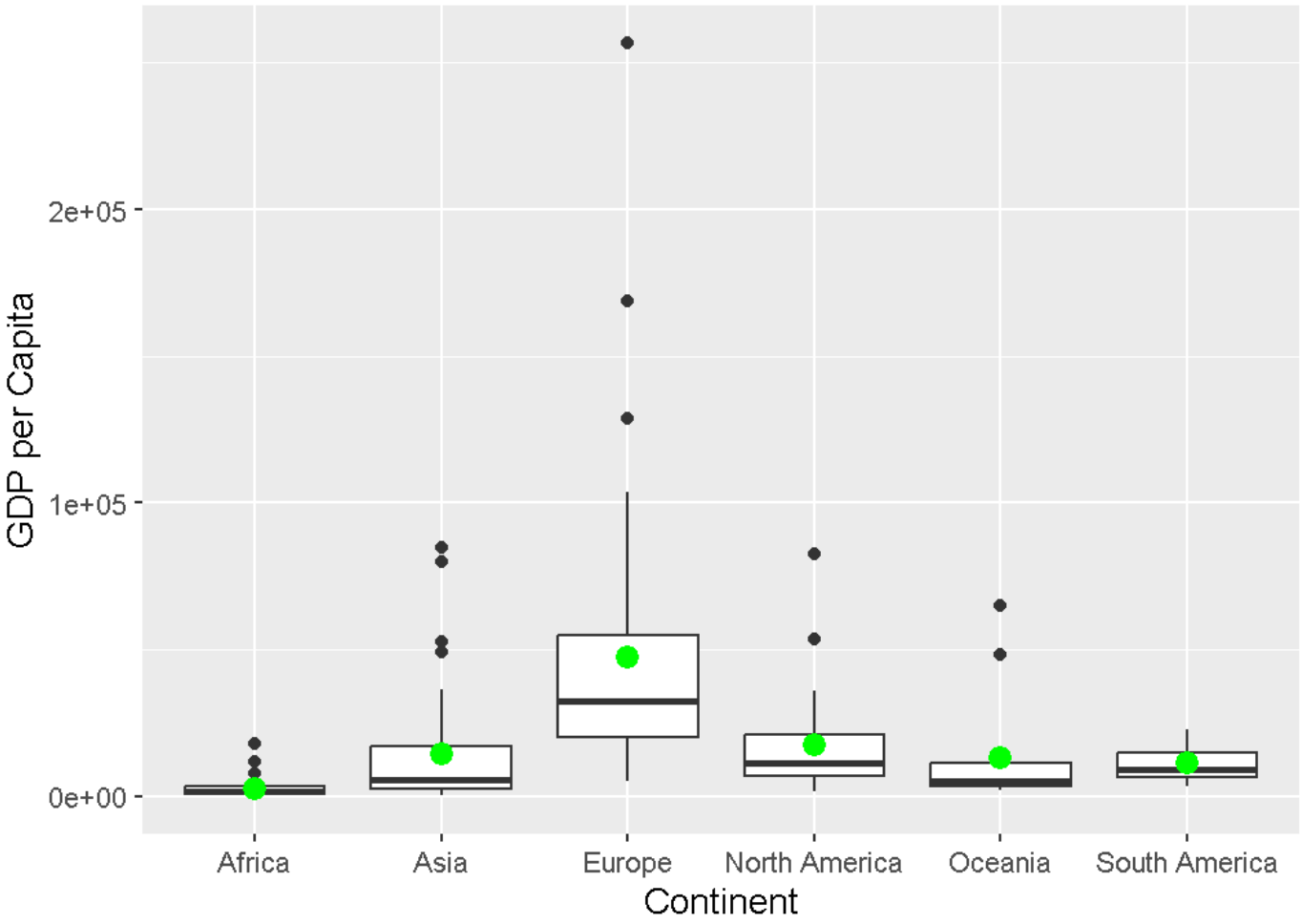


OUTLIERS/BOX PLOTS

Continents x GDP per Capita

Europe has the highest median and mean GDP per capita, but also the widest spread and many high-value outliers. Africa has the lowest GDP per capita, with a very small IQR and many low outliers. Asia and North America have moderate spreads, but North America has some very high outliers. Oceania and South America have relatively small spreads and lower means compared to Europe and North America.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = Continent, y = `GDP per Capita`))+
+  stat_summary(aes(x = Continent, y = `GDP per Capita`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```

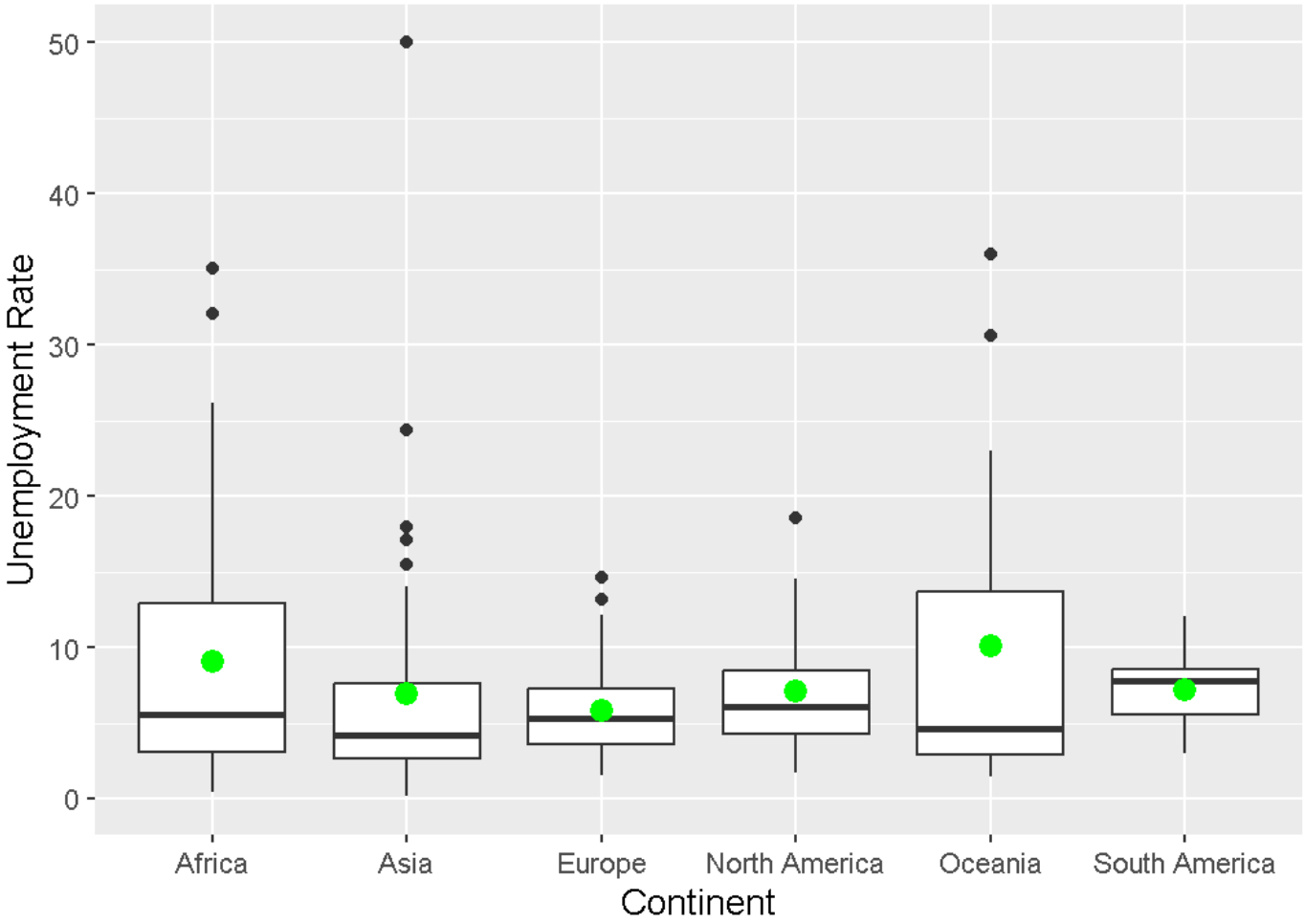


OUTLIERS/BOX PLOTS

Continents x Unemployment Rate

Africa and Oceania have the most variability in unemployment rates. Asia has the highest single outlier. Europe, South America, and North America are relatively stable. The presence of many outliers suggests significant economic disparities within continents.

```
> ggplot(edu_money)+
+  geom_boxplot(aes(x = Continent, y = `Unemployment Rate`))+
+  stat_summary(aes(x = Continent, y = `Unemployment Rate`), fun= mean,
+               geom = 'point', color = 'green', size = 3)
```



Cluster Analysis

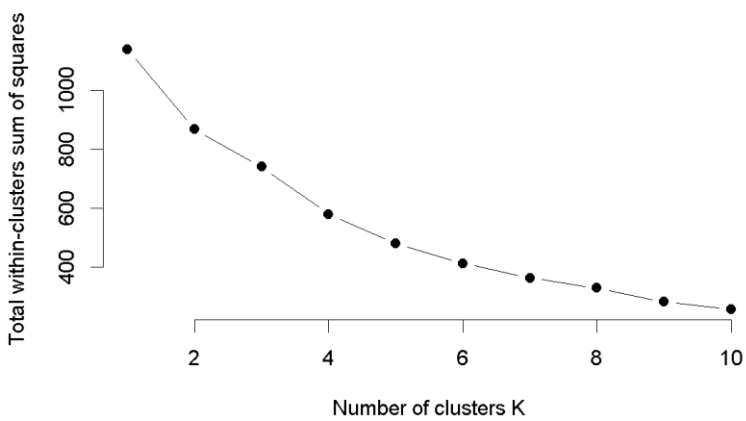
Kmeans

I chose to analyze with k-means clustering because it's efficient for large datasets (and this one had 191 countries), it helps uncover natural groupings without predefined labels, and it allows me to look at correlation and visualization analyses by adding a categorical dimension to numeric data.

The first step in performing a K-means analysis is using only numeric data. This new data set is called edu_money_var. The next step was to scale all my data. The third step is to find out how many clusters to use. After looking at the graph, the elbow appears to be around 3 or 4, and since I can categorize education into low, medium, and high, I am choosing to go with 3 clusters.

Step 1

```
> edu_money_var <- edu_money
> edu_money_var$Country <- NULL
> edu_money_var$`GDP per Capita Category` <- NULL
> edu_money_var$Continent <- NULL
Step 2
> edu_money_scaled <- scale(edu_money_var)
Step 3
> wss <- sapply(1:10, function(k){
+    kmeans(edu_money_scaled, centers = k, nstart = 10)$tot.withinss
+ })
> plot(1:10, wss, type = "b", pch = 19, frame = FALSE,
+     xlab = "Number of clusters K",
+     ylab = "Total within-clusters sum of squares")
>
```




```
Cluster means:
  Literacy Rate Physician Density        GDP GDP Growth GDP per Capita Unemployment
Rate
1    -1.9593028        -0.9915300 -0.2036685 -0.2122284     -0.5755541        -
0.2632431
2     0.2134876        -0.3710597 -0.1518755  0.2460202     -0.3540422
0.3065245
3     0.6717699         1.1825090  0.3759487 -0.3192238      0.9302496        -
0.3983614

Clustering vector:
[1] 1 2 2 3 2 2 3 2 3 3 3 2 2 2 2 3 3 2 1 1 2 2 2 2 2 3 1 1 2 2 2 2 3 1 1
```

```
 [36] 2 3 2 1 2 2 3 3 2 3 2 3 2 2 2 2 2 2 2 2 3 2 1 2 3 3 2 3 3 2 3 2 2 1 1
 [71] 2 1 2 3 3 2 2 2 2 3 3 3 2 3 2 2 2 2 3 2 2 3 2 2 1 2 3 3 3 1 1 2 3 1 3
[106] 2 1 2 2 2 3 3 2 2 2 1 2 2 2 2 3 3 2 1 1 3 2 3 2 1 2 2 2 1 2 2 2 3 3 3
[141] 3 3 2 2 2 2 2 3 3 1 2 2 1 3 3 3 1 1 2 3 1 3 2 1 2 3 3 2 2 2 2 1 1 2 3
[176] 2 2 2 2 2 3 3 3 3 2 2 2 2 1 2 2

Within cluster sum of squares by cluster:
[1]  98.25363 303.19928 343.59734
 (between_SS / total_SS =  34.6 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
"betweenss"
[7] "size"         "iter"         "ifault"
```

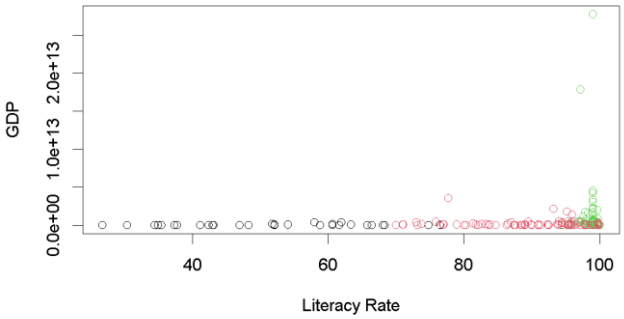
Cluster 1 is dominated by Very Low GDP countries, and has a few that fall in the low section. Cluster 2 spans mostly Low to Mid GDP, but dips into the very low and high columns. Cluster 3 has mostly High and Very High GDP countries.

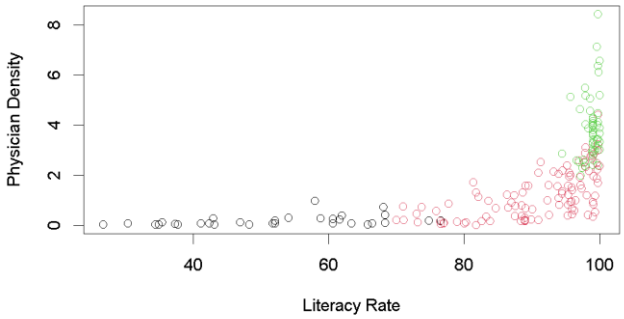
```
> table(edu_money$`GDP per Capita Category`, kmean_of_scaled$cluster)
```

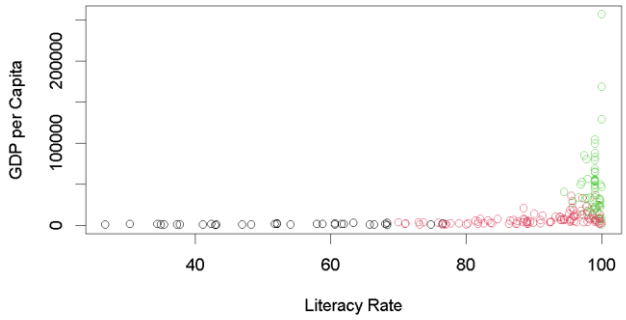
	Very low	Low	Mid	High	Very high
1	26	5	0	0	0
2	14	49	30	9	0
3	1	1	10	26	20

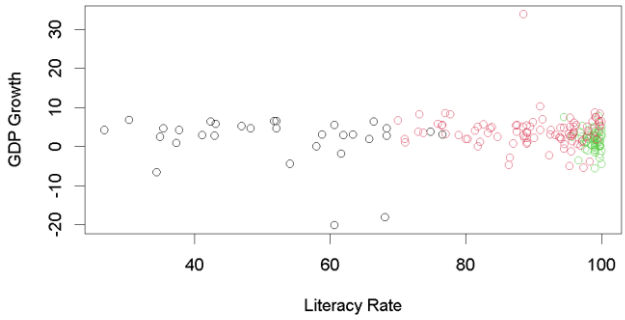
From the graphs it is easy to see that Cluster 1 countries are often in the lower-left (low literacy, low GDP), Cluster 2 is usually in the middle, and That Cluster 3 countries tend to group in the upper-right of plots meaning high literacy and higher GDP. It is ver clear in the first 9 graphs where most catigories are compared to literacy or Physician Density. The last group of graphs are more muddy and not easy to grab information out of.

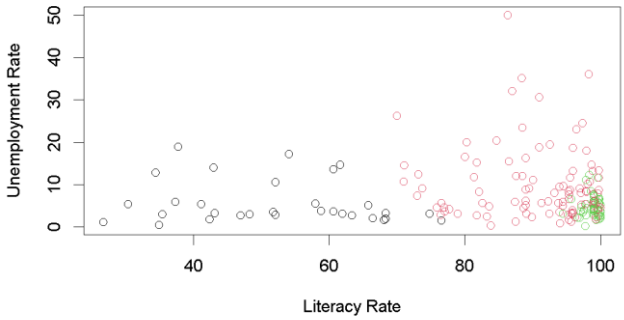
```
> plot(edu_money[c("Literacy Rate", "GDP per Capita")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("Literacy Rate", "Physician Density")], col =
kmean_of_scaled$cluster)
> plot(edu_money[c("Literacy Rate", "GDP")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("Literacy Rate", "Physician Density")], col =
kmean_of_scaled$cluster)
> plot(edu_money[c("Literacy Rate", "GDP Growth")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("Literacy Rate", "Unemployment Rate")], col =
kmean_of_scaled$cluster)
> plot(edu_money[c("Physician Density", "GDP")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("Physician Density", "GDP Growth")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("Physician Density", "GDP per Capita")], col =
kmean_of_scaled$cluster)
> plot(edu_money[c("Physician Density", "Unemployment Rate")],
colkmean_of_scaled$cluster)
> plot(edu_money[c("GDP", "GDP Growth")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("GDP", "GDP per Capita")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("GDP", "Unemployment Rate")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("GDP Growth", "GDP per Capita")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("GDP Growth", "Unemployment Rate")], col = kmean_of_scaled$cluster)
> plot(edu_money[c("GDP per Capita"), "Unemployment Rate")], col
=kmean_of_scaled$cluster) > plot(edu_money[c("GDP per Capita", "Unemployment Rate")],
col = kmean_of_scaled$cluster)
```

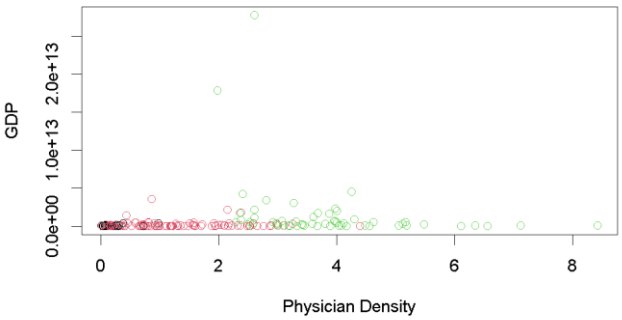

GDP
0.0e+00
1.0e+13
2.0e+13
40
60
80
100
Literacy Rate

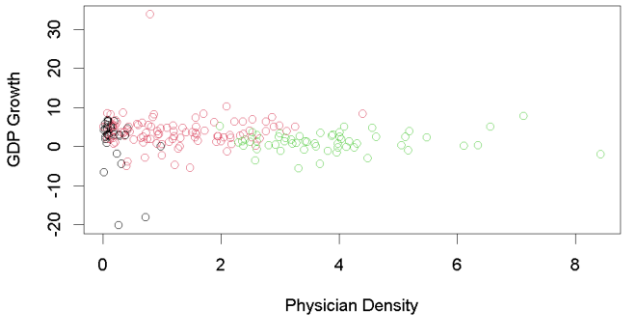

Physician Density
0
2
4
6
8
40
60
80
100
Literacy Rate

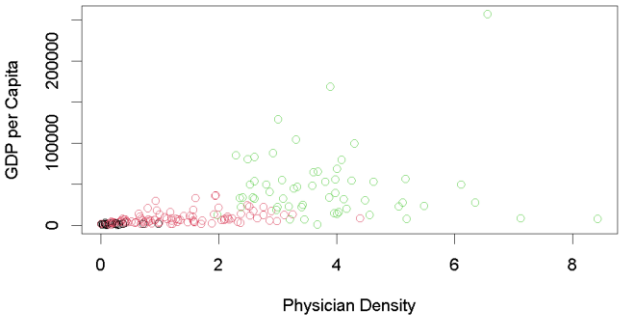

GDP per Capita
0
100000
200000
40
60
80
100
Literacy Rate

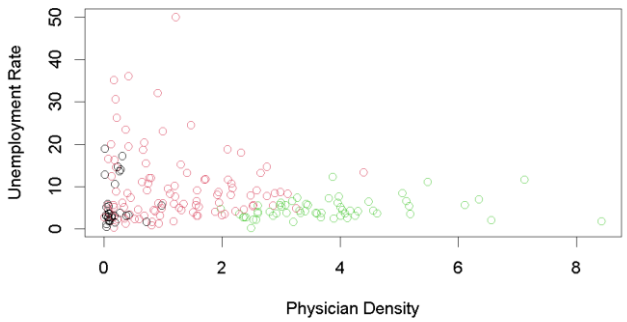

GDP Growth
-20
-10
0
10
20
30
40
60
80
100
Literacy Rate

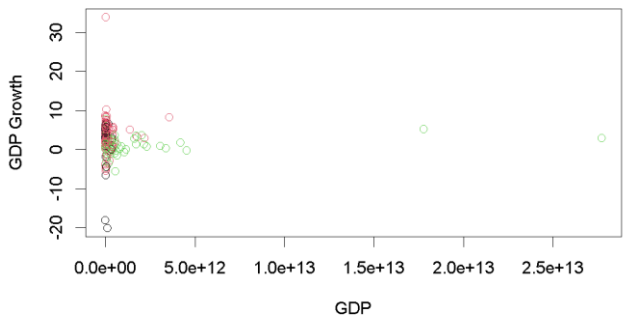

Unemployment Rate
0
10
20
30
40
50
40
60
80
100
Literacy Rate

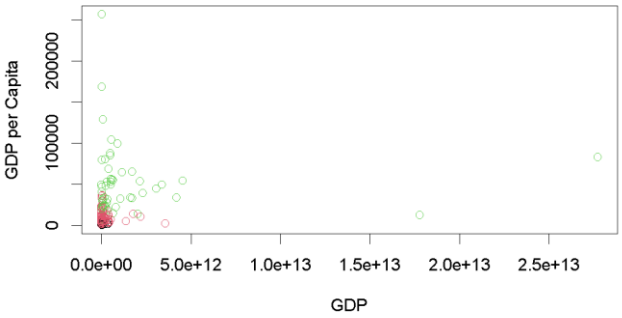

GDP
0.0e+00
1.0e+13
2.0e+13
0
2
4
6
8
Physician Density

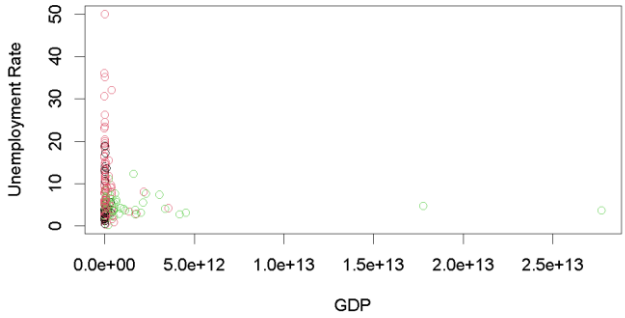

GDP Growth
-20
-10
0
10
20
30
0
2
4
6
8
Physician Density

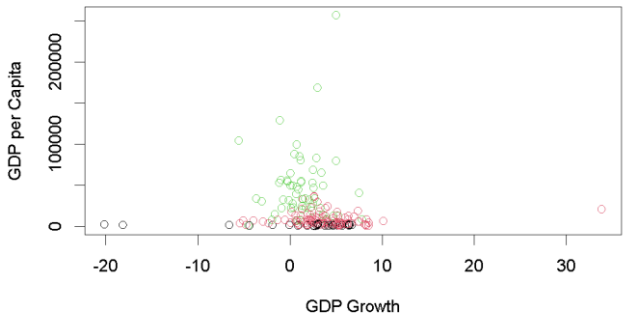

GDP per Capita
0
100000
200000
0
2
4
6
8
Physician Density

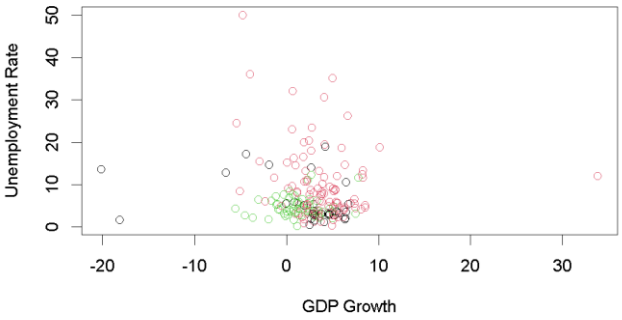

Unemployment Rate
0
10
20
30
40
50
0
2
4
6
8
Physician Density

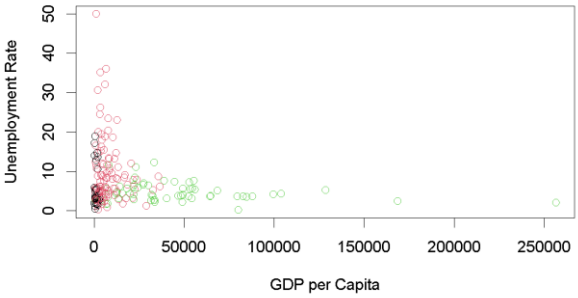

GDP Growth
-20
-10
0
10
20
30
0.0e+00
5.0e+12
1.0e+13
1.5e+13
2.0e+13
2.5e+13
GDP


GDP per Capita
0
100000
200000
0.0e+00
5.0e+12
1.0e+13
1.5e+13
2.0e+13
2.5e+13
GDP


Unemployment Rate
0
10
20
30
40
50
0.0e+00
5.0e+12
1.0e+13
1.5e+13
2.0e+13
2.5e+13
GDP


GDP per Capita
0
100000
200000
-20
-10
0
10
20
30
GDP Growth


Unemployment Rate
0
10
20
30
40
50
-20
-10
0
10
20
30
GDP Growth


Unemployment Rate
0
10
20
30
40
50
0
50000
100000
150000
200000
250000
GDP per Capita

Regression models (A)

In the first model I wanted to see if Higher literacy rates are associated with higher GDP per capital is supported. I made GDP per capital the response variable and literacy rate the predictor variable.

- Literacy Rate Coefficient is 634.9, which means for every 1 percent increase in literacy, GDP per capita increases by about $635 on average.
- The p-value is 1.86e-07. This is extremely small (< 0.001), so the relationship is statistically significant.
- R-squared is 0.1343, in other words about 13.4% of the variation in GDP per capita is explained by literacy rate alone.

```
> model1 <- lm(`GDP per Capita` ~ `Literacy Rate`, data = edu_money)
> summary(model1)

Call:
lm(formula = `GDP per Capita` ~ `Literacy Rate`, data = edu_money)

Residuals:
   Min     1Q Median     3Q    Max 
-26471 -15788  -8778   6553 229527 

Coefficients:
               Estimate Std. Error t value Pr(>|t|)    
(Intercept)     -36435.8    10408.5  -3.501 0.000579 ***
`Literacy Rate`    634.9      117.3   5.414 1.86e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 28490 on 189 degrees of freedom
Multiple R-squared:  0.1343,	Adjusted R-squared:  0.1297 
F-statistic: 29.31 on 1 and 189 DF,  p-value: 1.858e-07
```

REGRESSION MODELS (B)

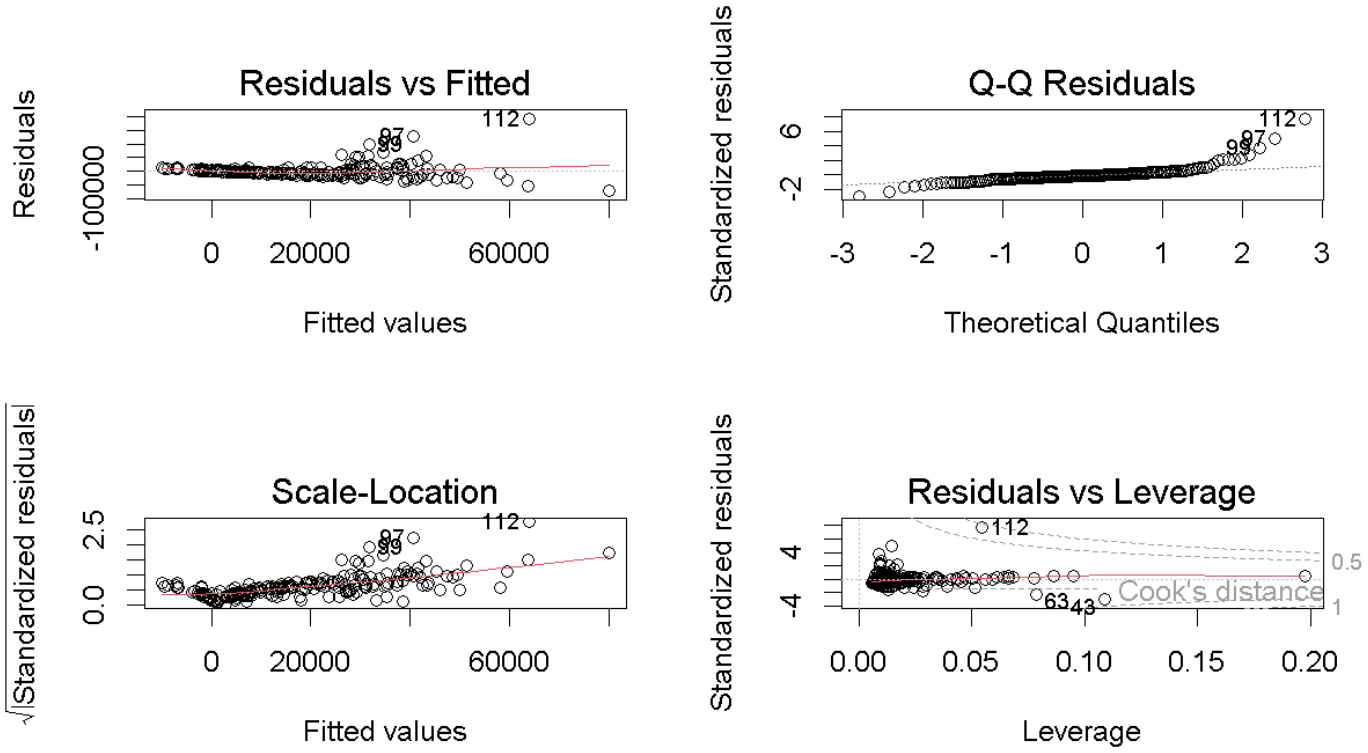
In this model, I added Physician Density and Unemployment Rate to see whether they would improve the prediction of GDP per Capita.

- The p-value for Literacy Rate is 0.3190, which is not statistically significant.
- For Tor Physician Density, the p-value is less than 0.001, indicating a highly significant difference. For every additional physician per 1,000 people, GDP per capita increases by an estimated 8,593 dollars.
- The Unemployment Rate has a p-value of 0.053. This is almost significant at the 0.05 level. Suggests that lower unemployment slightly increases GDP per capita.
- The R-squared value is 0.3086, meaning that approximately 31% of the variation in GDP per capita is explained by these three variables.

Adding Physician Density dramatically improved the model and is the strongest predictor. Although the effect of literacy rate disappears when healthcare and unemployment are taken into account.

```
> model2 <- lm(`GDP per Capita` ~ `Literacy Rate` + `Physician Density` + `Unemployment Rate`,
data = edu_money)
> summary(model2)
Call:
lm(formula = `GDP per Capita` ~ `Literacy Rate` + `Physician Density` +
    `Unemployment Rate`, data = edu_money)
Residuals:
   Min     1Q Median     3Q    Max
-72370  -9984  -2774   4457 192675
Coefficients:
                        Estimate/ Std. Error/ t value /Pr(>|t|)
(Intercept)          -4820.9  /  10426.0 / -0.462   /0.6443
`Literacy Rate`        134.0  /    134.1  / 0.999  / 0.3190
`Physician Density`   8593.2   / 1428.3  / 6.016 / 9.2e-09 ***
`Unemployment Rate`   -521.6  /    268.0 / -1.946 /  0.0531 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
Residual standard error: 25590 on 187 degrees of freedom
Multiple R-squared:  0.3086,  Adjusted R-squared:  0.2975
F-statistic: 27.82 on 3 and 187 DF,  p-value: 6.398e-15

par(mfrow = c(2, 2))
plot(model2)
```


Residuals vs Fitted
Residuals
Fitted values
-100000
0
20000
60000
112
97
99
Q-Q Residuals
Standardized residuals
Theoretical Quantiles
-2
6
-3
-2
-1
0
1
2
3
112
97
99
Scale-Location
√|Standardized residuals|
Fitted values
0.0
2.5
0
20000
60000
112
97
99
Residuals vs Leverage
Standardized residuals
Leverage
-4
4
0.00
0.05
0.10
0.15
0.20
112
63
43
Cook's distance
0.5
1

SUMMARY

I looked at how education and healthcare, measured by literacy rate and physician density, relate to economic performance in 191 countries. My main expectation was that countries with higher literacy and better access to doctors would have stronger economies, like higher GDP, faster growth, and lower unemployment.

General takeaways:

- The correlation matrix showed a moderate positive link between literacy rate and GDP per person at 0.37. In other words, countries with higher literacy rates usually have higher economic output per person. The boxplots backed this up, showing that literacy rates go up as GDP per capita increases. Countries in the 'Very High' GDP group had almost perfect literacy rates. The cluster analysis also grouped countries with high literacy rates together with those that have high GDP per capita. All of these points to education being linked to economic prosperity.
- There was a strong positive correlation of 0.54 between physician density and GDP per capita. This means that countries with more money usually have better access to doctors. When looking at the data, physician density goes up as GDP per capita increases, and you can see this clearly in the boxplots and heatmaps. Wealthier areas like Europe and North America have more doctors available. Overall, this shows that countries with more resources tend to invest more in healthcare.
- When looking at the contingency tables, it was clear that there were big differences between continents. Most African countries fell into the 'Very Low' and 'Low' GDP groups, while most European countries were in the 'High' and 'Very High' categories. The boxplots showed that Europe had the highest literacy rates and the most physicians, but Africa had the lowest numbers for both. The heatmaps made these gaps even more obvious, with darker colors in Africa and Asia showing lower literacy and less access to healthcare. All of these points highlight major differences in education, healthcare, and economic development within a clearly defined geographic area.
- When looking at clusters analysis, countries were grouped by their economic and social indicators, it found three main clusters. The countries with the highest GDP per capita also had higher literacy rates and more doctors, which matches the idea that education and healthcare help drive economic success.
- In the regression analysis, Literacy Rate is used to predict GDP per Capita in Model 1. The R^2 comes out to 0.13, which is a significant positive effect of literacy. Model 2 combines Literacy Rate, Physician Density, and Unemployment Rate to predict the GDP per Capita. R^2 came out to 0.31, which is a big jump from model 1. All in all, education correlates with economic

strength, but its effect diminishes when healthcare and labor factors are taken into account. Also, Physician density is a powerful predictor of GDP per capita, reinforcing the importance of healthcare infrastructure.

Overall, while the data supported the link between education and economic strength, it also highlighted the nuanced and multifactorial nature of economic growth.

The findings partially supported these expectations:

Hypothesis A:

Higher literacy rates are associated with higher GDP per capita.
Supported.

The data showed a moderate positive correlation of 0.37 between literacy rate and GDP per capita. In general, countries where more than 95% of people can read and write also had much higher GDP per person. This lines up with the idea that education helps boost a country's economy.

Hypothesis B:

GDP, GDP growth, and GDP per person are all correlated.
Not supported.

There was not much of a connection between GDP and GDP per person, and GDP growth did not really line up with either one. This means that even if a country is wealthier or has a higher GDP per person, its growth rate can still be very different. Other factors probably play a big role in how fast a country's economy grows.

Hypothesis C:

Countries with higher physician density have lower unemployment rates.

Slightly supported.

There was a small negative correlation of −0.17 between the number of doctors per person and the unemployment rate. In other words, countries with more access to healthcare could have had fewer people out of work. This supports the idea that investing in people's health can help improve job opportunities.

Hypothesis D:

Countries in the "Very High" GDP per capita category have significantly lower unemployment rates than those in the "Very Low" category.
Supported.

When looking at the data, countries with very high GDP per person tended to have lower and more consistent unemployment rates. In contrast, countries with very low GDP per person had much more variation, and more cases with higher unemployment were observed.

Follow-Up Questions

1. Why is it that countries can see their GDP go up, but not see the same kind of improvement in education and healthcare? What makes these things not always move together?
2. How does income inequality show up in places where the country as a whole is rich, but most people aren't seeing those benefits?
3. How do people's views about education in different countries end up shaping their economies?
4. Do some parts of the world see bigger payoffs from investing in education and healthcare than others? If so, what might explain those differences?
5. What kinds of policies have helped countries go from being less wealthy to having a much higher GDP per person?
6. What happens in a country when there are more doctors available? Does that connect to other things like life expectancy or quality of life?
7. How much does knowing how to use technology, and having access to it, actually help a country's economy grow?

REFERENCES

Mohamed, O. (2025). *How Education Drives Economic Growth*. Kaggle.com. https://www.kaggle.com/datasets/omarmohammed70/how-education-drives-economic-growth?resource=download

(2023). *Continents of the World by Population, Area & Density (2025 Data)*. https://worldostats.com/continents/